STANDARD OPERATING PROCEDURE

Cross-Flavivirus Convergent Host Response Analysis

Bioinformatics Execution Protocol v2.0

PHASE 0 — PROJECT SETUP

Step 0.1: Project Definition

Objective

Establish the scientific framework, datasets, and resources before any computation begins.

Central Hypothesis

DENV and ZIKV converge on a shared host transcriptomic response enriched for proviral host factors, partially mediated by suppression of shared regulatory miRNAs whose loss de-represses coordinated proviral gene programs.

Biological Model

```
Flavivirus infection
    → Suppression of specific host miRNAs
        → Loss of post-transcriptional repression of proviral genes
            → Coordinated upregulation of proviral gene modules
                → Cellular environment permissive for replication
```

Both DENV and ZIKV trigger this cascade through the same miRNA-gene-pathway regulatory network, explaining their convergent host transcriptomic response.

Dataset Architecture

Dataset	Accession	Role	Virus	Cell Type	Samples
Primary Discovery	GSE110496	Both DENV and ZIKV in same experiment	DENV-2 + ZIKV	Huh7 (human hepatoma)	~2100 cells
ZIKV Validation	GSE118305	Independent ZIKV immune cell data	ZIKV	Human monocyte-derived macrophages	60 samples
DENV Validation	GSE94892	Independent DENV patient data	DENV	Patient PBMCs	39 patients
Neural	GSE78711	ZIKV neural	ZIKV	iPSC-derived	8 samples

Dataset	Accession	Role	Virus	Cell Type	Samples
Extension		progenitor cell data		human NPCs	

Literature-Derived Resources

Resource	Description	Source
91 DENV-downregulated miRNAs	Non-redundant set from two studies	Ouyang et al., Li et al.
55 cross-flavivirus miRNAs	Subset with predicted ZIKV genome binding	miRNAProtPred analysis
Curated host factors	Proviral and antiviral genes from literature	Multiple publications

Software Requirements

Software	Version	Purpose
R	>= 4.2	All statistical analyses
Seurat	>= 5.0	Single-cell analysis
DESeq2	>= 1.38	Differential expression
clusterProfiler	>= 4.6	Pathway enrichment
ReactomePA	>= 1.42	Reactome enrichment
org.Hs.eg.db	current	Gene annotation
multiMiR	>= 1.20	miRNA target prediction
SuperExactTest	>= 1.0	Multi-set intersection testing
UpSetR	>= 1.4	Intersection visualization
ggplot2	>= 3.4	Plotting
pheatmap or ComplexHeatmap	current	Heatmaps
Cytoscape	>= 3.10	Network visualization
STRINGdb	>= 2.10	PPI data
cytoHubba	current	Hub gene identification (Cytoscape plugin)
MCODE	current	Module detection (Cytoscape plugin)

Directory Structure

```
project/
├── 00_raw_data/
│   ├── GSE110496/
│   ├── GSE118305/
```

```
│   ├── GSE94892/
│   └── GSE78711/
├── 01_processed_data/
│   ├── seurat_objects/
│   ├── pseudobulk/
│   └── deg_tables/
├── 02_literature_resources/
│   ├── host_factors_curated.csv
│   ├── denv_downregulated_mirnas.csv
│   └── mirna_target_predictions.csv
├── 03_results/
│   ├── phase3_shared_degs/
│   ├── phase4_host_factors/
│   ├── phase5_mirna/
│   ├── phase6_pathways/
│   ├── phase7_validation/
│   └── phase8_network/
├── 04_figures/
│   ├── main/
│   └── supplementary/
├── 05_tables/
│   ├── main/
│   └── supplementary/
└── scripts/
    ├── phase1_acquisition.R
    ├── phase2_singlecell.R
    ├── phase3_degs.R
    ├── phase4_shared.R
    ├── phase5_hostfactors.R
    ├── phase6_mirna.R
    ├── phase7_pathways.R
    ├── phase8_validation.R
    └── phase9_network.R
```

PHASE 1 — DATASET ACQUISITION AND VERIFICATION

Step 1.1: Acquire GSE110496 (Discovery Dataset)

Objective

Download the primary dataset containing DENV-2 and ZIKV single-cell transcriptomic data.

Input Files

- GEO accession: GSE110496
- Publication: Zanini et al., eLife 2018 (DOI: 10.7554/eLife.32942)

Pre-Analysis Checks Before Download

1. Navigate to https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE110496
2. Confirm: Organism = Homo sapiens

3. Confirm: Platform = Illumina NextSeq 500
4. Confirm: Experiment type includes "Expression profiling"
5. Check supplementary files for processed count matrices
6. ALSO check eLife paper supplementary materials: https://elifesciences.org/articles/32942 — the authors may have deposited processed data as supplementary tables or in a GitHub repository

Analysis Procedure

```
# Attempt 1: Download from GEO
library(GEOquery)

# Check if Series Matrix files exist
gse <- getGEO("GSE110496", GSEMatrix = TRUE, getGPL = FALSE)

# If this returns expression data: extract and examine
if (length(gse) > 0) {
  expr_data <- exprs(gse[[1]])
  meta_data <- pData(gse[[1]])
  cat("Dimensions:", dim(expr_data), "\n")
  cat("Metadata columns:", colnames(meta_data), "\n")
}

# Attempt 2: Download supplementary files
getGEOSuppFiles("GSE110496", baseDir = "00_raw_data/GSE110496/")
# List what was downloaded
list.files("00_raw_data/GSE110496/GSE110496/", recursive = TRUE)

# Attempt 3: Check SRA for raw FASTQ
# Go to: https://www.ncbi.nlm.nih.gov/sra?term=GSE110496
# Note the SRP accession number
# Use SRA Toolkit if raw data processing is needed

# Attempt 4: Check the Zanini lab GitHub
# https://github.com/iosonofabio
# The lab may have deposited processed data
```

Quality Control Checks After Download

- ☐ Expression matrix contains human gene names (HGNC symbols or Ensembl IDs)
- ☐ Metadata contains columns for: virus type (DENV/ZIKV/Control), MOI, timepoint (4h/12h/24h/48h)
- ☐ Viral RNA read counts per cell are available (essential for viral load correlation analysis)
- ☐ Total cell count matches expected range (~2100 sequenced cells per Zanini et al.)
- ☐ No condition has zero cells
- ☐ Gene names are consistent (no mixed Ensembl/Symbol formats)

Output Files

- `00_raw_data/GSE110496/expression_matrix.csv` (or .rds, .h5ad depending on format)

- `00_raw_data/GSE110496/cell_metadata.csv`
- `00_raw_data/GSE110496/data_acquisition_log.txt` (record: date downloaded, source, format, any issues)

Interpretation

This step produces raw data only. No biological conclusions.

Go / No-Go Decision

CONTINUE if: Expression matrix and metadata with virus/timepoint labels are obtained. **INVESTIGATE** if: Data downloads but metadata is incomplete. Check the eLife supplementary tables. Contact corresponding author if necessary. **STOP** if: Data cannot be obtained in any usable format after all attempts. The project cannot proceed as designed without this dataset.

Things That Must Be Checked Before Proceeding

- ☐ Cell count per condition documented
- ☐ Virus labels verified (not just "treated"/"untreated" but specifically DENV/ZIKV/Control)
- ☐ Timepoint labels verified
- ☐ Gene count per cell distribution examined (histogram)
- ☐ Data format compatible with Seurat (sparse matrix or count matrix)

Step 1.2: Acquire GSE118305 (ZIKV Validation Dataset)

Objective

Download the independent ZIKV macrophage dataset for validation.

Input Files

- GEO accession: GSE118305
- Publication: Carlin et al., 2018 (PMID: 30206152)

Pre-Analysis Checks Before Download

1. Navigate to https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE118305
2. Confirm: 60 samples listed
3. Confirm: Organism = Homo sapiens
4. Note: This dataset contains BOTH RNA-seq and ChIP-seq data. Use ONLY the RNA-seq samples.
5. Sample naming convention: "HumanX_RNA_[Condition]_[Timepoint]"
    - Mock = uninfected control

- FSS_ZIKVpositive = ZIKV-infected (sorted by intracellular viral staining)
- FSS_ZIKVnegative = bystander (exposed but not infected)
- SD001_ZIKVpositive = second ZIKV strain, infected
- SD001_ZIKVnegative = second strain, bystander

Analysis Procedure

```
library(GEOquery)

gse118305 <- getGEO("GSE118305", GSEMatrix = TRUE, getGPL = FALSE)

# Examine metadata to identify RNA-seq samples
meta <- pData(gse118305[[1]])
print(colnames(meta))

# Identify RNA-seq vs ChIP-seq samples
# RNA-seq samples have "RNA" in their title
rna_samples <- grep("RNA", meta$title, value = FALSE)

# Download supplementary files (likely contains FPKM table)
getGEOSuppFiles("GSE118305", baseDir = "00_raw_data/GSE118305/")

# The supplementary file likely named something like:
# "GSE118305_RNAseq_HMDM_ZIKV_2018.txt"
# This should contain FPKM or raw counts for all RNA-seq samples
```

Key Sample Selection for This Project

For the validation analysis, use:

- **Test group**: ZIKV-positive samples (FSS strain, 24h timepoint) — cells confirmed to contain ZIKV
- **Control group**: Mock samples (24h timepoint) — uninfected controls
- Do NOT use ZIKV-negative (bystander) cells as controls — they were exposed to virus and may have paracrine transcriptomic changes

Quality Control Checks After Download

- ☐ RNA-seq samples identified and separated from ChIP-seq
- ☐ At least 3 biological replicates (different human donors) per comparison group
- ☐ FPKM or count values are present (not just peak files)
- ☐ Sample labels allow unambiguous assignment to ZIKV+, ZIKV-, Mock groups
- ☐ Donor identity is traceable (Human1, Human3, Human4)

Output Files

- `00_raw_data/GSE118305/expression_matrix_rnaseq.csv`
- `00_raw_data/GSE118305/sample_metadata.csv`
- `00_raw_data/GSE118305/data_acquisition_log.txt`

Go / No-Go Decision

CONTINUE if: Expression data for ZIKV+ and Mock samples with at least 3 donors is obtained. **INVESTIGATE** if: Only FPKM provided (not raw counts). FPKM can be used with limma-voom but not with DESeq2. If raw counts are needed, download from SRA. **STOP** if: Fewer than 2 biological replicates per group. The dataset cannot support proper differential expression analysis.

Things That Must Be Checked Before Proceeding

- ☐ Number of RNA-seq samples documented
- ☐ Comparison groups defined (ZIKV+ vs. Mock at 24h)
- ☐ Donor replication confirmed
- ☐ Data format confirmed (counts vs. FPKM)

Step 1.3: Acquire GSE94892 (DENV Validation Dataset)

Objective

Download the DENV patient PBMC dataset for validation.

Input Files

- GEO accession: GSE94892
- Publication: Suppiah et al., 2017 (PMID: 28683259)

Pre-Analysis Checks Before Download

1. Navigate to GEO page
2. Confirm: RNA-seq from human PBMCs
3. Note the severity classification system (WHO 2009 criteria)
4. Determine whether raw counts or processed expression values are deposited

Analysis Procedure

```
library(GEOquery)

gse94892 <- getGEO("GSE94892", GSEMatrix = TRUE, getGPL = FALSE)
meta <- pData(gse94892[[1]])

# Identify sample groups
# Expected: DWS (dengue without warning signs), DWWS, SD (severe dengue),
controls
print(table(meta$characteristics_ch1))

# Download supplementary files
getGEOSuppFiles("GSE94892", baseDir = "00_raw_data/GSE94892/")
```

Key Sample Selection

- **Test group**: All DENV-infected patients (pool DWS, DWWS, SD into single "DENV-infected" group for primary analysis)
- **Control group**: Controls (other febrile illnesses or healthy)
- **Secondary analysis**: Stratify by severity (optional, for supplementary materials)

Quality Control Checks After Download

- ☐ Total patient count >= 30 (expected: 39)
- ☐ Clear DENV+ vs. control labels
- ☐ Gene identifiers are standard (HGNC symbols or Ensembl)
- ☐ No samples with anomalously low read counts or gene counts

Output Files

- `00_raw_data/GSE94892/expression_matrix.csv`
- `00_raw_data/GSE94892/patient_metadata.csv`
- `00_raw_data/GSE94892/data_acquisition_log.txt`

Go / No-Go Decision

CONTINUE if: Expression data with >= 25 patients and clear DENV/control labels. **INVESTIGATE** if: Only normalized data available (use limma for analysis instead of DESeq2). **STOP** if: Fewer than 15 patients or no control samples.

Things That Must Be Checked Before Proceeding

- ☐ Patient count per group documented
- ☐ Severity classification recorded
- ☐ Data format confirmed

Step 1.4: Acquire GSE78711 (Neural Extension Dataset)

Objective

Download the ZIKV neural progenitor cell dataset. This dataset tests whether the shared DENV-ZIKV host response identified in Huh7 cells and validated in immune cells extends to neural tissue, the primary site of ZIKV neuropathogenesis.

Input Files

- GEO accession: GSE78711
- Publication: Tang et al., Cell Stem Cell 2016 (DOI: 10.1016/j.stem.2016.02.016)

Pre-Analysis Checks Before Download

1. Navigate to https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE78711
2. Confirm: Organism = Homo sapiens
3. Confirm: Cell type = iPSC-derived human neural progenitor cells (hNPCs)
4. Confirm: Virus = ZIKV (strain MR766, African lineage)
5. Note: Sample size is small (expected 8 samples: ZIKV-infected and mock in duplicate)

Analysis Procedure

```
library(GEOquery)

gse78711 <- getGEO("GSE78711", GSEMatrix = TRUE, getGPL = FALSE)
meta <- pData(gse78711[[1]])

# Examine sample structure
print(meta[, c("title", "source_name_ch1", "characteristics_ch1")])
cat("Total samples:", nrow(meta), "\n")

# Download supplementary files (may contain count matrix or FPKM)
getGEOSuppFiles("GSE78711", baseDir = "00_raw_data/GSE78711/")
list.files("00_raw_data/GSE78711/GSE78711/", recursive = TRUE)
```

Key Information About This Dataset

- **Cell type:** iPSC-derived human neural progenitor cells — the cell type most relevant to ZIKV-induced microcephaly
- **ZIKV strain:** MR766 (African lineage). Note: this is the prototype laboratory strain, not the Asian lineage (PRVABC59) responsible for the 2015-2016 epidemic. Some transcriptomic differences between lineages have been reported. Acknowledge this in the manuscript.
- **Sample size:** Small (expected ~4 ZIKV-infected + ~4 mock control). This limits statistical power for independent DEG discovery but is adequate for testing whether specific genes from the discovery analysis are also perturbed in neural cells.
- **Role in this project:** This dataset is NOT used for primary discovery or for DENV-ZIKV comparison (DENV does not infect neural progenitor cells). It is used exclusively to test whether the shared DENV-ZIKV host response extends beyond hepatoma and immune cells into neural cells.

Quality Control Checks After Download

- ☐ Expression data obtained (counts or FPKM)
- ☐ ZIKV-infected and Mock control samples clearly identified
- ☐ At least 2 replicates per condition
- ☐ Gene identifiers are standard (HGNC symbols, Ensembl, or RefSeq)

Output Files

- `00_raw_data/GSE78711/expression_matrix.csv`

- `00_raw_data/GSE78711/sample_metadata.csv`
- `00_raw_data/GSE78711/data_acquisition_log.txt`

Go / No-Go Decision

CONTINUE if: Expression data for ZIKV-infected and mock NPC samples obtained with clear labels. **INVESTIGATE** if: Only processed data (FPKM) available. Use limma for differential expression. **EXCLUDE FROM NEURAL EXTENSION** if: Data cannot be downloaded or has fewer than 2 replicates per condition. The main study (Phases 3-8) is unaffected. Only the neural extension analysis (Phase 8, Step 8.4) is lost.

Things That Must Be Checked Before Proceeding

- ☐ Expression data format documented (counts vs. FPKM vs. TPM)
- ☐ Sample count per condition documented
- ☐ ZIKV strain noted (MR766 — African lineage)
- ☐ Data acquisition logged

Step 1.5: Prepare Literature-Derived Host Factor Dataset

Objective

Create an analysis-ready table of curated proviral and antiviral host genes.

Input Files

- Your independently curated host factor gene list (from published literature)
- HGNC symbol checker: https://www.genenames.org/tools/multi-symbol-checker/

Pre-Analysis Checks

1. All gene symbols must be current HGNC approved symbols
2. No duplicate entries
3. Each gene must have a clear proviral or antiviral classification
4. Each classification must have at least one PubMed reference

Analysis Procedure

Step 1: Export your curated list into a spreadsheet with these exact columns:

Column Name	Type	Description	Example
gene_symbol	string	Official HGNC symbol	BCL2L1
entrez_id	integer	NCBI Entrez Gene ID	598
classification	string	"proviral" or "antiviral"	proviral

Column Name	Type	Description	Example
virus_evidence	string	Supporting virus(es)	DENV, ZIKV, JEV
mechanism	string	Functional description	Anti-apoptotic; sustains infected cells
pathway	string	Associated pathway	Apoptosis / BCL-2 family
evidence_type	string	Experimental method	siRNA knockdown reduces viral titer
confidence	string	"high" or "moderate"	high
pmid	string	PubMed ID(s)	30123456; 31234567

Step 2: Validate gene symbols

```
# Submit all gene symbols to HGNC Multi-Symbol Checker
# https://www.genenames.org/tools/multi-symbol-checker/
# Record which symbols were:
#   - Approved (no change needed)
#   - Previous symbol (update to current)
#   - Not found (investigate manually)

# In R, also verify using org.Hs.eg.db
library(org.Hs.eg.db)
library(clusterProfiler)

host_factors <- read.csv("02_literature_resources/host_factors_curated.csv")

# Check which symbols map to Entrez IDs
mapped <- bitr(host_factors$gene_symbol, fromType = "SYMBOL",
               toType = "ENTREZID", OrgDb = org.Hs.eg.db)

# Identify unmapped symbols
unmapped <- setdiff(host_factors$gene_symbol, mapped$SYMBOL)
cat("Unmapped symbols:", length(unmapped), "\n")
if (length(unmapped) > 0) {
  cat("These need manual verification:\n")
  print(unmapped)
}
```

Step 3: Remove duplicates and finalize

```
# Check for duplicates
dups <- host_factors$gene_symbol[duplicated(host_factors$gene_symbol)]
if (length(dups) > 0) {
  cat("DUPLICATE ENTRIES found:", dups, "\n")
  cat("Resolve manually: keep the entry with stronger evidence\n")
}

# Save final version
write.csv(host_factors, "02_literature_resources/host_factors_curated.csv",
row.names = FALSE)
```

Quality Control Checks

- ☐ All gene symbols are current HGNC approved symbols

- ☐ Zero duplicate gene entries
- ☐ Every gene has a classification (no blanks)
- ☐ Every gene has at least one PMID
- ☐ Proviral gene count >= 15 (required for enrichment testing power)
- ☐ Antiviral gene count >= 10
- ☐ Total gene count >= 30 (ideally 50+)
- ☐ No gene appears in BOTH proviral and antiviral lists (conflicting classification must be resolved)

Output Files

- `02_literature_resources/host_factors_curated.csv` — final curated table
- `02_literature_resources/host_factors_proviral.txt` — one gene symbol per line, proviral only
- `02_literature_resources/host_factors_antiviral.txt` — one gene symbol per line, antiviral only
- `02_literature_resources/host_factor_validation_report.txt` — symbol mapping results

Go / No-Go Decision

CONTINUE if: >= 30 total genes, >= 15 proviral, all symbols validated. **INVESTIGATE** if: 20-29 genes total. Enrichment tests may be underpowered. Consider expanding curation. **STOP** if: < 20 genes total. The enrichment analysis will not have statistical power.

Things That Must Be Checked Before Proceeding

- ☐ Final gene count documented (total, proviral, antiviral)
- ☐ All symbols validated
- ☐ No duplicates
- ☐ No conflicting classifications

Step 1.6: Prepare DENV-Downregulated miRNA Dataset

Objective

Create an analysis-ready table of the 91 non-redundant DENV-downregulated miRNAs and define the three analysis sets (55, 91, 36).

Input Files

- Your compiled miRNA list from Ouyang et al. and Li et al.
- miRNAProtPred results (Supplementary File 2 from current manuscript)
- miRBase: https://www.mirbase.org/ for ID validation

Analysis Procedure

Step 1: Create master miRNA table

Column Name	**Type**	**Description**
mirna_id	string	miRBase mature ID (e.g., hsa-miR-134-5p)
source_study	string	"Ouyang" or "Li" or "Both"
log2FC	numeric	Fold change from original study
binds_ZIKV_genome	boolean	TRUE if in the 55-miRNA set
n_ZIKV_binding_sites	integer	Number of binding sites from miRNAProtPred
best_MFE	numeric	Best minimum free energy (kcal/mol)

Step 2: Define the three analysis sets

```
mirnas <- read.csv("02_literature_resources/denv_downregulated_mirnas.csv")

# Set 1: 55 cross-flavivirus miRNAs (PRIMARY)
set_55 <- mirnas$mirna_id[mirnas$binds_ZIKV_genome == TRUE]

# Set 2: All 91 DENV-downregulated miRNAs (SENSITIVITY)
set_91 <- mirnas$mirna_id

# Set 3: 36 DENV-only miRNAs without ZIKV binding (SPECIFICITY CONTROL)
set_36 <- mirnas$mirna_id[mirnas$binds_ZIKV_genome == FALSE]

# Verify counts
cat("Set 55 (primary):", length(set_55), "\n")        # Should be 55
cat("Set 91 (sensitivity):", length(set_91), "\n")     # Should be 91
cat("Set 36 (specificity):", length(set_36), "\n")     # Should be 36
cat("Check: 55 + 36 =", length(set_55) + length(set_36), "\n")  # Should be 91

# Save
writeLines(set_55, "02_literature_resources/mirna_set_55_primary.txt")
writeLines(set_91, "02_literature_resources/mirna_set_91_sensitivity.txt")
writeLines(set_36, "02_literature_resources/mirna_set_36_specificity.txt")
```

Step 3: Validate miRNA IDs against miRBase

```
# Check that all miRNA IDs are current miRBase entries
# Some IDs may have changed between miRBase versions
# Manual check: search each ID at https://www.mirbase.org/
# Common issues:
#   - hsa-miR-XXX vs hsa-miR-XXX-3p vs hsa-miR-XXX-5p (arm specification)
#   - Deprecated IDs from older miRBase versions
```

Quality Control Checks

- ☐ Exactly 91 unique miRNAs in the full set
- ☐ Exactly 55 in the ZIKV-binding set
- ☐ 55 + 36 = 91 (sets are complementary, no overlap)
- ☐ All miRNA IDs match current miRBase nomenclature
- ☐ No duplicate entries

Output Files

- `02_literature_resources/denv_downregulated_mirnas.csv` — master table
- `02_literature_resources/mirna_set_55_primary.txt` — primary set
- `02_literature_resources/mirna_set_91_sensitivity.txt` — sensitivity set
- `02_literature_resources/mirna_set_36_specificity.txt` — specificity control

Step 1.7: Generate miRNA Target Predictions

Objective

Predict the host mRNA targets of all 91 DENV-downregulated miRNAs using multiple databases.

Input Files

- `02_literature_resources/denv_downregulated_mirnas.csv`

Pre-Analysis Checks

- ☐ multiMiR R package installed and functional
- ☐ Internet connection available (multiMiR queries external databases)
- ☐ miRNA IDs are in the format accepted by multiMiR (hsa-miR-XXX-Xp)

Analysis Procedure

```
library(multiMiR)

mirnas_all <- scan("02_literature_resources/mirna_set_91_sensitivity.txt",
                    what = "character")

# Query predicted targets from multiple databases
# This may take 10-30 minutes depending on server load
predicted <- get_multimir(mirna = mirnas_all,
                           table = "predicted",
                           summary = TRUE,
                           predicted.cutoff = 35,
```

```
                           predicted.cutoff.type = "p")

# Query experimentally validated targets
validated <- get_multimir(mirna = mirnas_all,
                           table = "validated",
                           summary = TRUE)

# Extract data frames
pred_df <- as.data.frame(predicted@data)
val_df <- as.data.frame(validated@data)

# CRITICAL: Filter for high-confidence predictions
# Require prediction in at least 2 independent databases
target_db_count <- table(pred_df$target_symbol, pred_df$database)
high_conf_genes <- names(which(rowSums(target_db_count > 0) >= 2))

cat("Total unique predicted targets:", length(unique(pred_df$target_symbol)),
"\n")
cat("High-confidence targets (2+ databases):", length(high_conf_genes), "\n")
cat("Validated targets:", length(unique(val_df$target_symbol)), "\n")

# Save complete target mapping
write.csv(pred_df, "02_literature_resources/mirna_target_predictions_all.csv",
          row.names = FALSE)
write.csv(val_df, "02_literature_resources/mirna_target_validated.csv",
          row.names = FALSE)
writeLines(high_conf_genes,
"02_literature_resources/mirna_targets_high_conf.txt")

# Create per-set target lists
mirnas_55 <- scan("02_literature_resources/mirna_set_55_primary.txt",
                   what = "character")
mirnas_36 <- scan("02_literature_resources/mirna_set_36_specificity.txt",
                   what = "character")

targets_55 <- unique(pred_df$target_symbol[pred_df$mature_mirna_id %in%
mirnas_55])
targets_36 <- unique(pred_df$target_symbol[pred_df$mature_mirna_id %in%
mirnas_36])

# Apply same 2-database filter per set
pred_55 <- pred_df[pred_df$mature_mirna_id %in% mirnas_55, ]
tc_55 <- table(pred_55$target_symbol, pred_55$database)
hc_targets_55 <- names(which(rowSums(tc_55 > 0) >= 2))

pred_36 <- pred_df[pred_df$mature_mirna_id %in% mirnas_36, ]
tc_36 <- table(pred_36$target_symbol, pred_36$database)
hc_targets_36 <- names(which(rowSums(tc_36 > 0) >= 2))

writeLines(hc_targets_55,
"02_literature_resources/mirna_targets_55_highconf.txt")
writeLines(hc_targets_36,
"02_literature_resources/mirna_targets_36_highconf.txt")

cat("High-conf targets of 55 miRNAs:", length(hc_targets_55), "\n")
cat("High-conf targets of 36 miRNAs:", length(hc_targets_36), "\n")
```

Quality Control Checks

- ☐ multiMiR returned results for at least 80 of the 91 miRNAs (some may not be in all databases)

- ☐ Total unique predicted targets >= 3000 for the full 91-miRNA set
- ☐ High-confidence targets (2+ databases) >= 1500
- ☐ No miRNA returned zero targets (investigate if so — ID mismatch likely)
- ☐ Target lists for 55 and 36 sets are non-empty

Output Files

- `02_literature_resources/mirna_target_predictions_all.csv` — complete prediction table
- `02_literature_resources/mirna_target_validated.csv` — validated targets only
- `02_literature_resources/mirna_targets_high_conf.txt` — all 91 miRNAs, high-confidence targets
- `02_literature_resources/mirna_targets_55_highconf.txt` — 55-miRNA set, high-confidence targets
- `02_literature_resources/mirna_targets_36_highconf.txt` — 36-miRNA set, high-confidence targets

Go / No-Go Decision

CONTINUE if: >= 1500 high-confidence targets across the 91-miRNA set. **INVESTIGATE** if: < 1000 targets. Check miRNA ID formatting. Some databases may be temporarily unavailable. **REPEAT** if: multiMiR query timed out or returned errors. Try again with smaller batches of miRNAs.

Things That Must Be Checked Before Proceeding to Phase 2

- ☐ All datasets downloaded and verified (Steps 1.1-1.4)
- ☐ GSE110496 (discovery): expression matrix and metadata acquired
- ☐ GSE118305 (ZIKV validation): expression data and sample labels confirmed
- ☐ GSE94892 (DENV validation): expression data and patient metadata confirmed
- ☐ GSE78711 (neural extension): expression data acquired, ZIKV strain noted as MR766
- ☐ Host factor list prepared and validated (Step 1.5)
- ☐ miRNA sets defined (55, 91, 36) and validated (Step 1.6)
- ☐ miRNA target predictions generated for all three sets (Step 1.7)
- ☐ All output files exist in the expected directories

PHASE 2 — SINGLE-CELL PROCESSING (GSE110496)

Step 2.1: Load and Inspect Data

Objective

Load the GSE110496 expression data into Seurat and characterize the dataset structure.

Input Files

- Expression matrix from Step 1.1
- Cell metadata from Step 1.1

Pre-Analysis Checks

- ☐ Expression matrix is a genes x cells matrix (rows = genes, columns = cells)
- ☐ If matrix is cells x genes: transpose it
- ☐ Values appear to be integer counts (not normalized values). Check: `summary(as.numeric(expr_matrix[1:100, 1:100]))`
- ☐ Metadata has one row per cell, matching column names of expression matrix

Analysis Procedure

```
library(Seurat)
library(ggplot2)

# Load data — adjust path and function based on data format
# Option A: CSV/TSV count matrix
counts <- read.csv("00_raw_data/GSE110496/expression_matrix.csv",
                    row.names = 1, check.names = FALSE)
meta <- read.csv("00_raw_data/GSE110496/cell_metadata.csv",
                  row.names = 1)

# Option B: If data is in H5AD format (from Python)
# library(SeuratDisk)
# Convert("00_raw_data/GSE110496/data.h5ad", dest = "h5seurat")
# sobj <- LoadH5Seurat("00_raw_data/GSE110496/data.h5seurat")

# Create Seurat object
sobj <- CreateSeuratObject(counts = counts, meta.data = meta,
                            project = "FlaviConv")

# Verify metadata columns
cat("Metadata columns:", colnames(sobj@meta.data), "\n")

# CRITICAL: Identify and standardize the condition column
# The metadata must contain a column distinguishing DENV/ZIKV/Control
# This may be named: "condition", "infection", "virus", "treatment", etc.
# Standardize to "condition" with values: "DENV", "ZIKV", "Control"

# Check what exists
print(table(sobj@meta.data$condition))  # adjust column name

# Similarly identify the timepoint column
# Standardize to "timepoint" with values: "4h", "12h", "24h", "48h"
print(table(sobj@meta.data$timepoint))  # adjust column name

# Check for MOI information
print(table(sobj@meta.data$MOI))  # if exists
```

```
# If viral read counts exist, record the column name
# Zanini et al. quantified viral RNA per cell
# This may be in a column like "viral_reads" or "vRNA"
```

Quality Control Checks

- ☐ Total cells: record exact number
- ☐ Cells per condition: document (expect ~300-700 per condition per timepoint)
- ☐ Cells per timepoint: document
- ☐ Gene count: total genes in matrix (expect 15,000-25,000 for human)
- ☐ Median genes per cell: expect 2000-6000 for Smart-seq2
- ☐ Median UMIs per cell: expect higher for Smart-seq2 than 10X
- ☐ Viral read column identified (Y/N)

Output Files

- `01_processed_data/seurat_objects/sobj_raw.rds` — raw Seurat object before QC
- `03_results/phase2_qc/cell_count_summary.csv` — cells per condition per timepoint

Things That Must Be Checked Before Proceeding

- ☐ Condition labels are correct and complete (every cell labeled)
- ☐ Timepoint labels are correct and complete
- ☐ No ambiguity in sample assignment

Step 2.2: Quality Control Filtering

Objective

Remove low-quality cells and potential doublets to ensure reliable downstream analysis.

Input Files

- `01_processed_data/seurat_objects/sobj_raw.rds`

Analysis Procedure

```
sobj <- readRDS("01_processed_data/seurat_objects/sobj_raw.rds")

# Calculate QC metrics
sobj[["percent.mt"]] <- PercentageFeatureSet(sobj, pattern = "^MT-")
sobj[["percent.ribo"]] <- PercentageFeatureSet(sobj, pattern = "^RP[SL]")

# Visualize QC metrics BEFORE filtering
```

```
VlnPlot(sobj, features = c("nFeature_RNA", "nCount_RNA", "percent.mt"),
        group.by = "condition", pt.size = 0.1)
ggsave("04_figures/supplementary/QC_violin_prefilter.pdf", width = 12, height =
4)

# Scatter plot: genes vs counts (to identify doublets)
FeatureScatter(sobj, feature1 = "nCount_RNA", feature2 = "nFeature_RNA",
               group.by = "condition")
ggsave("04_figures/supplementary/QC_scatter_prefilter.pdf", width = 8, height =
6)

# DETERMINE THRESHOLDS
# These depend on the sequencing technology:
# Smart-seq2 (plate-based): higher gene counts per cell (2000-8000)
# 10X Chromium (droplet-based): lower gene counts (500-5000)
# GSE110496 uses a custom Smart-seq2-like protocol

# Apply thresholds — adjust based on pre-filter distributions
sobj_filtered <- subset(sobj,
  subset = nFeature_RNA > 2000 &    # minimum genes per cell
           nFeature_RNA < 8000 &    # maximum genes (doublet filter)
           percent.mt < 15          # maximum mitochondrial %
)

# Document filtering results
cat("Cells before filtering:", ncol(sobj), "\n")
cat("Cells after filtering:", ncol(sobj_filtered), "\n")
cat("Cells removed:", ncol(sobj) - ncol(sobj_filtered), "\n")
cat("Removal rate:", round(100 * (1 - ncol(sobj_filtered)/ncol(sobj)), 1), "%
\n")

# CRITICAL CHECK: no condition lost all its cells
print(table(sobj_filtered$condition, sobj_filtered$timepoint))
```

Quality Control Checks

- ☐ Removal rate < 30% (if > 30%: thresholds may be too stringent, re-examine distributions)
- ☐ Every condition x timepoint combination retains >= 30 cells
- ☐ No systematic bias: one condition does not lose disproportionately more cells than others
- ☐ Mitochondrial percentage distribution looks normal (peak around 5-10%, no condition with median > 15%)

Output Files

- `01_processed_data/seurat_objects/sobj_filtered.rds`
- `03_results/phase2_qc/filtering_summary.csv` — cells before/after per condition
- `04_figures/supplementary/QC_violin_prefilter.pdf`
- `04_figures/supplementary/QC_violin_postfilter.pdf`

Step 2.3: Normalization and Dimensionality Reduction

Objective

Normalize expression values, identify variable genes, and generate low-dimensional embeddings for visualization.

Input Files

- `01_processed_data/seurat_objects/sobj_filtered.rds`

Analysis Procedure

```
sobj <- readRDS("01_processed_data/seurat_objects/sobj_filtered.rds")

# Normalize
sobj <- NormalizeData(sobj, normalization.method = "LogNormalize",
                       scale.factor = 10000)

# Find variable features
sobj <- FindVariableFeatures(sobj, selection.method = "vst",
                              nfeatures = 3000)

# Scale data (regress out mitochondrial percentage)
sobj <- ScaleData(sobj, vars.to.regress = "percent.mt")

# PCA
sobj <- RunPCA(sobj, npcs = 50)

# Determine significant PCs
ElbowPlot(sobj, ndims = 50)
ggsave("04_figures/supplementary/elbow_plot.pdf", width = 6, height = 4)
# Choose number of PCs: use the elbow point (typically 15-30)

# UMAP
n_pcs <- 20  # adjust based on elbow plot
sobj <- RunUMAP(sobj, dims = 1:n_pcs)

# Visualizations
p1 <- DimPlot(sobj, group.by = "condition",
               cols = c("Control" = "grey70", "DENV" = "#E41A1C", "ZIKV" =
"#377EB8"))
p2 <- DimPlot(sobj, group.by = "timepoint")
p1 + p2
ggsave("04_figures/supplementary/UMAP_condition_timepoint.pdf", width = 14,
height = 6)

# If viral load column exists, visualize
if ("viral_reads" %in% colnames(sobj@meta.data)) {
  FeaturePlot(sobj, features = "viral_reads")
  ggsave("04_figures/supplementary/UMAP_viral_load.pdf", width = 7, height = 6)
}

# Save
saveRDS(sobj, "01_processed_data/seurat_objects/sobj_processed.rds")
```

Quality Control Checks

- ☐ PCA shows separation between infected and control cells on PC1 or PC2 (especially at later timepoints)

- ☐ UMAP shows some structure by condition (complete overlap of all conditions would suggest very weak infection effect)
- ☐ No condition clusters entirely apart due to batch effects rather than biology
- ☐ Variable features include known infection-response genes (check for ISGs: ISG15, MX1, OAS1, IFIT1)

Output Files

- `01_processed_data/seurat_objects/sobj_processed.rds` — fully processed Seurat object
- `04_figures/supplementary/elbow_plot.pdf`
- `04_figures/supplementary/UMAP_condition_timepoint.pdf`
- `04_figures/supplementary/UMAP_viral_load.pdf`

Things That Must Be Checked Before Proceeding to Phase 3

- ☐ Seurat object is fully processed (normalized, scaled, PCA, UMAP)
- ☐ Cell count per condition per timepoint documented
- ☐ UMAP shows biologically interpretable structure
- ☐ Viral load column identified (if available)

PHASE 3 — DIFFERENTIAL EXPRESSION ANALYSIS

Step 3.1: Pseudo-Bulk Aggregation and Replicate Assessment

Objective

Aggregate single-cell data into pseudo-bulk samples and determine whether biological replicates exist for proper statistical inference.

Input Files

- `01_processed_data/seurat_objects/sobj_processed.rds`

Pre-Analysis Checks

THIS IS THE MOST IMPORTANT PRE-CHECK IN THE ENTIRE PROJECT

Before running DESeq2, you MUST determine whether biological replicates exist within each condition at each timepoint. Pseudo-bulk DEA with DESeq2 requires sample-level replication to estimate gene-level dispersion. Without replicates, DESeq2 p-values are unreliable.

Check the metadata:

```
sobj <- readRDS("01_processed_data/seurat_objects/sobj_processed.rds")

# Check for replicate information
cat("Metadata columns:\n")
print(colnames(sobj@meta.data))

# Look for columns like: "replicate", "batch", "experiment", "well", "plate"
# Zanini et al. performed a replicate experiment for DENV at 1/5 scale
# Check if this replicate is annotated

# Count unique replicates per condition per timepoint
if ("replicate" %in% colnames(sobj@meta.data)) {
  print(table(sobj$condition, sobj$timepoint, sobj$replicate))
}
```

Analysis Procedure — Two Paths

PATH A: Biological replicates exist (>= 2 per condition per timepoint)

```
# Pseudo-bulk: aggregate by condition x timepoint x replicate
# Each combination becomes one pseudo-bulk sample

library(DESeq2)

# At 24h timepoint (primary)
sobj_24h <- subset(sobj, timepoint == "24h")

pb_24h <- AggregateExpression(sobj_24h,
  group.by = c("condition", "replicate"),
  return.seurat = FALSE,
  slot = "counts")$RNA

# Create sample info
# Columns of pb_24h should be named like: "Control_rep1", "DENV_rep1", etc.
sample_names <- colnames(pb_24h)
sample_info <- data.frame(
  condition = gsub("_rep.*", "", sample_names),
  replicate = gsub(".*_", "", sample_names),
  row.names = sample_names
)

dds_24h <- DESeqDataSetFromMatrix(
  countData = round(pb_24h),  # ensure integers
  colData = sample_info,
  design = ~ condition
)

dds_24h <- DESeq(dds_24h)

# DEGs for DENV vs Control
res_denv_24h <- results(dds_24h, contrast = c("condition", "DENV", "Control"))
# DEGs for ZIKV vs Control
res_zikv_24h <- results(dds_24h, contrast = c("condition", "ZIKV", "Control"))
```

PATH B: No biological replicates exist (common for single-cell experiments)

Use the cross-timepoint combined model as the primary statistical analysis:

```
# Pseudo-bulk: aggregate by condition x timepoint
# Each combination becomes one pseudo-bulk sample
# 3 conditions x 4 timepoints = 12 pseudo-bulk samples

pb_all <- AggregateExpression(sobj,
```

```
  group.by = c("condition", "timepoint"),
  return.seurat = FALSE,
  slot = "counts")$RNA

sample_names <- colnames(pb_all)
# Parse condition and timepoint from column names
sample_info <- data.frame(
  condition = sub("_.*", "", sample_names),
  timepoint = sub(".*_", "", sample_names),
  row.names = sample_names
)

# Convert timepoint to ordered factor
sample_info$timepoint <- factor(sample_info$timepoint,
  levels = c("4h", "12h", "24h", "48h"))

dds_all <- DESeqDataSetFromMatrix(
  countData = round(pb_all),
  colData = sample_info,
  design = ~ timepoint + condition
)

dds_all <- DESeq(dds_all)

# DEGs (averaged across timepoints, controlling for temporal variation)
res_denv_combined <- results(dds_all, contrast = c("condition", "DENV",
"Control"))
res_zikv_combined <- results(dds_all, contrast = c("condition", "ZIKV",
"Control"))
```

ALSO generate 24h-specific fold changes using Seurat (for visualization):

```
# Single-cell level DEGs at 24h (for fold-change values, not p-values)
sobj_24h <- subset(sobj, timepoint == "24h")

markers_denv_24h <- FindMarkers(sobj_24h,
  ident.1 = "DENV", ident.2 = "Control",
  group.by = "condition",
  test.use = "wilcox",
  logfc.threshold = 0, # get all genes
  min.pct = 0.1)

markers_zikv_24h <- FindMarkers(sobj_24h,
  ident.1 = "ZIKV", ident.2 = "Control",
  group.by = "condition",
  test.use = "wilcox",
  logfc.threshold = 0,
  min.pct = 0.1)
```

Choosing Which Results to Use

Result Source	**Use For**	**Reason**
DESeq2 combined model (Path B)	Formal p-values, adjusted p-values	Proper statistical framework with replication
Seurat FindMarkers at 24h	Log2FC values for scatter plot, Venn diagrams	Biologically interpretable at a specific timepoint
DESeq2 Path A (if replicates exist)	Both p-values and fold changes	Best of both worlds

Quality Control Checks

- ☐ Pseudo-bulk samples have >= 500,000 total counts each (very low counts indicate too few cells were aggregated)
- ☐ DESeq2 ran without errors
- ☐ dispersion estimates converged (check: `plotDispEsts(dds)`)
- ☐ MA plots show expected pattern: most genes near zero, with spread at higher expression levels
- ☐ Number of significant DEGs is biologically reasonable (see table below)

Expected DEG ranges (|log2FC| >= 1, padj < 0.05):

Comparison	Expected Range	If Lower	If Higher
DENV vs Control (24h)	100-2000	May be underpowered; try relaxed thresholds	Normal for active infection
ZIKV vs Control (24h)	100-2000	Same	Same
Combined model (any)	200-3000	Expected: combined has more power	Check for confounders

Output Files

- `01_processed_data/deg_tables/DEGs_DENV_vs_Control_combined.csv`
- `01_processed_data/deg_tables/DEGs_ZIKV_vs_Control_combined.csv`
- `01_processed_data/deg_tables/DEGs_DENV_vs_Control_24h_FC.csv` (fold changes from FindMarkers)
- `01_processed_data/deg_tables/DEGs_ZIKV_vs_Control_24h_FC.csv`
- `01_processed_data/pseudobulk/pseudobulk_counts.csv`
- `01_processed_data/pseudobulk/sample_info.csv`
- `04_figures/supplementary/dispersion_plot.pdf`
- `04_figures/supplementary/MA_plot_DENV.pdf`
- `04_figures/supplementary/MA_plot_ZIKV.pdf`

Go / No-Go Decision (GATE G1)

CONTINUE if: >= 200 significant DEGs per virus comparison. **INVESTIGATE** if: 50-199 DEGs. Try relaxed thresholds (|log2FC| >= 0.585, padj < 0.1). **STOP** if: < 50 DEGs for both viruses after threshold relaxation. The dataset does not yield sufficient signal for downstream analysis.

Things That Must Be Checked Before Proceeding

- ☐ Replicate structure documented (Path A or B)
- ☐ DEG counts per comparison documented

- ☐ Dispersion plot examined
- ☐ Fold-change distributions examined (histograms)
- ☐ DEG tables saved with all columns (gene, log2FC, pvalue, padj, baseMean)

PHASE 4 — SHARED DEG DISCOVERY

Step 4.1: Identify Shared DEGs

Objective

Determine which host genes are significantly perturbed by BOTH DENV and ZIKV.

Input Files

- `01_processed_data/deg_tables/DEGs_DENV_vs_Control_combined.csv`
- `01_processed_data/deg_tables/DEGs_ZIKV_vs_Control_combined.csv`

Pre-Analysis Checks

- ☐ Both DEG tables use the same gene identifiers (same column name, same format)
- ☐ Both tables contain padj column (adjusted p-values)
- ☐ log2FoldChange column has both positive and negative values

Analysis Procedure

```
# Load DEG tables
deg_denv <-
read.csv("01_processed_data/deg_tables/DEGs_DENV_vs_Control_combined.csv")
deg_zikv <-
read.csv("01_processed_data/deg_tables/DEGs_ZIKV_vs_Control_combined.csv")

# Define significance thresholds
FC_THRESHOLD <- 1        # |log2FC| >= 1
P_THRESHOLD <- 0.05      # padj < 0.05

# Classify DEGs
denv_up <- deg_denv$gene[deg_denv$log2FoldChange >= FC_THRESHOLD &
                          !is.na(deg_denv$padj) & deg_denv$padj < P_THRESHOLD]
denv_down <- deg_denv$gene[deg_denv$log2FoldChange <= -FC_THRESHOLD &
                            !is.na(deg_denv$padj) & deg_denv$padj < P_THRESHOLD]

zikv_up <- deg_zikv$gene[deg_zikv$log2FoldChange >= FC_THRESHOLD &
                          !is.na(deg_zikv$padj) & deg_zikv$padj < P_THRESHOLD]
zikv_down <- deg_zikv$gene[deg_zikv$log2FoldChange <= -FC_THRESHOLD &
                            !is.na(deg_zikv$padj) & deg_zikv$padj < P_THRESHOLD]

# Shared DEGs (concordant direction only)
shared_up <- intersect(denv_up, zikv_up)
shared_down <- intersect(denv_down, zikv_down)
shared_all <- c(shared_up, shared_down)
```

```
# Discordant genes (up in one, down in other)
discordant <- c(
  intersect(denv_up, zikv_down),
  intersect(denv_down, zikv_up)
)

# Virus-specific
denv_only <- setdiff(c(denv_up, denv_down), c(zikv_up, zikv_down))
zikv_only <- setdiff(c(zikv_up, zikv_down), c(denv_up, denv_down))

# Summary
summary_df <- data.frame(
  Category = c("DENV_up", "DENV_down", "ZIKV_up", "ZIKV_down",
               "Shared_up", "Shared_down", "Shared_total",
               "Discordant", "DENV_only", "ZIKV_only"),
  Count = c(length(denv_up), length(denv_down),
            length(zikv_up), length(zikv_down),
            length(shared_up), length(shared_down), length(shared_all),
            length(discordant), length(denv_only), length(zikv_only))
)
print(summary_df)
write.csv(summary_df, "03_results/phase3_shared_degs/deg_summary.csv", row.names
= FALSE)

# Save shared gene lists
write.csv(data.frame(gene = shared_up, direction = "up"),
          "03_results/phase3_shared_degs/shared_DEGs_up.csv", row.names = FALSE)
write.csv(data.frame(gene = shared_down, direction = "down"),
          "03_results/phase3_shared_degs/shared_DEGs_down.csv", row.names =
FALSE)
write.csv(data.frame(gene = shared_all,
                      direction = ifelse(shared_all %in% shared_up, "up",
"down")),
          "03_results/phase3_shared_degs/shared_DEGs_all.csv", row.names =
FALSE)
```

Quality Control Checks

- ☐ Shared_up + Shared_down = Shared_total (arithmetic check)
- ☐ Discordant genes are a small fraction of total shared + discordant (< 20% ideally; if > 50%, the viruses are producing opposing effects)
- ☐ Shared genes are not dominated by a single gene family or chromosome (check for artifacts)

Step 4.2: Statistical Significance of DEG Overlap (GATE G2)

Objective

Test whether the shared DEG overlap exceeds chance expectation.

Input Files

- DEG lists from Step 4.1
- Total measured genes (background) from DEG tables

Analysis Procedure

```
# Background: genes measured in BOTH comparisons
background_genes <- intersect(deg_denv$gene, deg_zikv$gene)
N <- length(background_genes)  # background size

# For UPREGULATED genes
n_denv_up <- length(denv_up)
n_zikv_up <- length(zikv_up)
n_shared_up <- length(shared_up)
expected_up <- (n_denv_up * n_zikv_up) / N

fisher_up <- fisher.test(matrix(c(
  n_shared_up,
  n_denv_up - n_shared_up,
  n_zikv_up - n_shared_up,
  N - n_denv_up - n_zikv_up + n_shared_up
), nrow = 2), alternative = "greater")

cat("=== UPREGULATED GENES ===\n")
cat("DENV up:", n_denv_up, "\n")
cat("ZIKV up:", n_zikv_up, "\n")
cat("Shared up:", n_shared_up, "\n")
cat("Expected by chance:", round(expected_up, 1), "\n")
cat("Fold enrichment:", round(n_shared_up / expected_up, 2), "\n")
cat("Fisher p-value:", fisher_up$p.value, "\n")
cat("Odds ratio:", round(fisher_up$estimate, 2), "\n\n")

# Repeat for DOWNREGULATED genes
# (same structure, using denv_down, zikv_down, shared_down)

# Repeat for ALL DEGs combined
# (same structure, using all DENV DEGs and all ZIKV DEGs)

# Save results
overlap_stats <- data.frame(
  Direction = c("Up", "Down", "All"),
  DENV_DEGs = c(n_denv_up, length(denv_down), n_denv_up + length(denv_down)),
  ZIKV_DEGs = c(n_zikv_up, length(zikv_down), n_zikv_up + length(zikv_down)),
  Shared = c(n_shared_up, length(shared_down), length(shared_all)),
  Expected = c(round(expected_up, 1), NA, NA),  # fill in
  Fold_Enrichment = c(round(n_shared_up / expected_up, 2), NA, NA),
  Fisher_p = c(fisher_up$p.value, NA, NA),
  Odds_Ratio = c(round(fisher_up$estimate, 2), NA, NA)
)
write.csv(overlap_stats, "03_results/phase3_shared_degs/overlap_statistics.csv",
          row.names = FALSE)
```

Venn Diagram

```
library(VennDiagram)
# or use ggVennDiagram or UpSetR

# Simple Venn for up + down combined
venn_data <- list(
  DENV = c(denv_up, denv_down),
  ZIKV = c(zikv_up, zikv_down)
)

# Using UpSetR for more detailed view
library(UpSetR)
all_genes <- unique(c(denv_up, denv_down, zikv_up, zikv_down))
upset_df <- data.frame(
```

```
  gene = all_genes,
  DENV_up = as.integer(all_genes %in% denv_up),
  DENV_down = as.integer(all_genes %in% denv_down),
  ZIKV_up = as.integer(all_genes %in% zikv_up),
  ZIKV_down = as.integer(all_genes %in% zikv_down)
)
pdf("04_figures/main/Figure2C_UpSet_DEGs.pdf", width = 8, height = 5)
upset(upset_df[, -1], order.by = "freq", sets.bar.color = "#4575B4")
dev.off()
```

Go / No-Go Decision (GATE G2)

CONTINUE if: Fisher $p < 0.001$ AND fold enrichment > 2 for upregulated genes. **INVESTIGATE** if: $p < 0.05$ but fold enrichment < 2. The overlap is significant but modest. Proceed cautiously. **CONSIDER PIVOTING** if: $p > 0.05$. The overlap is not significant. Try relaxed DEG thresholds ($|log2FC| >= 0.585$). If still not significant: proceed to Step 4.3 (fold-change correlation may reveal convergence not captured by binary DEG overlap). **STOP** if: $p > 0.05$ AND fold-change correlation (Step 4.3) is < 0.2. The convergence hypothesis is not supported.

Step 4.3: Fold-Change Correlation Analysis (GATE G3)

Objective

Quantify the genome-wide degree of host response conservation between DENV and ZIKV. This produces the central figure of the paper.

Input Files

- `01_processed_data/deg_tables/DEGs_DENV_vs_Control_24h_FC.csv` (24h fold changes)
- `01_processed_data/deg_tables/DEGs_ZIKV_vs_Control_24h_FC.csv`
- `02_literature_resources/host_factors_curated.csv`
- `02_literature_resources/mirna_targets_55_highconf.txt`

Pre-Analysis Checks

- ☐ Both DEG tables have a log2FoldChange (or avg_log2FC for Seurat) column
- ☐ Gene identifiers match between the two tables
- ☐ Host factor table and miRNA target list use the same gene identifiers

Analysis Procedure

```
library(ggplot2)

# Load 24h fold changes
fc_denv <-
read.csv("01_processed_data/deg_tables/DEGs_DENV_vs_Control_24h_FC.csv")
fc_zikv <-
read.csv("01_processed_data/deg_tables/DEGs_ZIKV_vs_Control_24h_FC.csv")
```

```
# Merge on gene name
merged <- merge(
  fc_denv[, c("gene", "avg_log2FC")],
  fc_zikv[, c("gene", "avg_log2FC")],
  by = "gene", suffixes = c("_DENV", "_ZIKV")
)

cat("Genes in both comparisons:", nrow(merged), "\n")

# Compute correlations
pearson <- cor.test(merged$avg_log2FC_DENV, merged$avg_log2FC_ZIKV,
                     method = "pearson")
spearman <- cor.test(merged$avg_log2FC_DENV, merged$avg_log2FC_ZIKV,
                      method = "spearman")

cat("Pearson r:", round(pearson$estimate, 4), "p:", pearson$p.value, "\n")
cat("Spearman rho:", round(spearman$estimate, 4), "p:", spearman$p.value, "\n")

# Add host factor and miRNA target annotations
host_factors <- read.csv("02_literature_resources/host_factors_curated.csv")
mirna_targets <- scan("02_literature_resources/mirna_targets_55_highconf.txt",
                       what = "character")

merged$category <- "Other"
merged$category[merged$gene %in%
  host_factors$gene_symbol[host_factors$classification == "proviral"]] <-
"Proviral"
merged$category[merged$gene %in%
  host_factors$gene_symbol[host_factors$classification == "antiviral"]] <-
"Antiviral"
merged$category[merged$gene %in% mirna_targets &
  merged$category == "Other"] <- "miRNA target"

# Factor ordering for plot layering
merged$category <- factor(merged$category,
  levels = c("Other", "miRNA target", "Antiviral", "Proviral"))

# THE CENTRAL FIGURE
p <- ggplot(merged, aes(x = avg_log2FC_DENV, y = avg_log2FC_ZIKV)) +
  # Background points (Other genes)
  geom_point(data = subset(merged, category == "Other"),
             color = "grey85", alpha = 0.3, size = 0.5) +
  # miRNA targets
  geom_point(data = subset(merged, category == "miRNA target"),
             color = "#FEB24C", alpha = 0.5, size = 1.2) +
  # Antiviral
  geom_point(data = subset(merged, category == "Antiviral"),
             color = "#4575B4", alpha = 0.8, size = 2.5) +
  # Proviral
  geom_point(data = subset(merged, category == "Proviral"),
             color = "#D73027", alpha = 0.8, size = 2.5) +
  # Diagonal (perfect conservation line)
  geom_abline(slope = 1, intercept = 0, linetype = "dashed",
              color = "grey50", linewidth = 0.3) +
  # Regression line
  geom_smooth(data = merged, method = "lm", color = "black",
              se = TRUE, linewidth = 0.5, alpha = 0.2) +
  # Labels
  labs(x = expression("log"[2]*"FC (DENV vs Control)"),
       y = expression("log"[2]*"FC (ZIKV vs Control)"),
       subtitle = paste0("Pearson r = ", round(pearson$estimate, 3),
                         ", p = ", format(pearson$p.value, digits = 3))) +
  # Theme
  theme_minimal(base_size = 12) +
```

```
  theme(panel.grid.minor = element_blank(),
        legend.position = "right") +
  coord_fixed(ratio = 1)

ggsave("04_figures/main/Figure2D_FoldChange_Correlation.pdf",
       p, width = 8, height = 7)

# Save merged table
write.csv(merged,
"03_results/phase3_shared_degs/merged_foldchanges_annotated.csv",
          row.names = FALSE)

# Save correlation results
cor_results <- data.frame(
  Test = c("Pearson", "Spearman"),
  Estimate = c(pearson$estimate, spearman$estimate),
  P_value = c(pearson$p.value, spearman$p.value),
  CI_lower = c(pearson$conf.int[1], NA),
  CI_upper = c(pearson$conf.int[2], NA)
)
write.csv(cor_results, "03_results/phase3_shared_degs/correlation_results.csv",
          row.names = FALSE)
```

Quality Control Checks

- ☐ Scatter plot shows a visible positive trend (not a random cloud)
- ☐ Proviral genes (red) are preferentially in the upper-right quadrant (both viruses upregulate them)
- ☐ The regression line has a positive slope close to the diagonal
- ☐ No extreme outliers dominating the correlation (check: remove top/bottom 1% and recompute)

Go / No-Go Decision (GATE G3)

STRONG SIGNAL if: Pearson r > 0.5. Proceed with full confidence to Phases 5-9. **MODERATE SIGNAL** if: Pearson r 0.3-0.5. Proceed but note as moderate convergence. **WEAK SIGNAL** if: Pearson r 0.2-0.3. Proceed cautiously. Focus on pathway-level rather than gene-level convergence. **NO SIGNAL** if: Pearson r < 0.2. The convergence hypothesis is not supported at the genome-wide level. The project must be restructured.

Output Files

- `04_figures/main/Figure2D_FoldChange_Correlation.pdf` — THE CENTRAL FIGURE
- `03_results/phase3_shared_degs/merged_foldchanges_annotated.csv`
- `03_results/phase3_shared_degs/correlation_results.csv`

Step 4.4: Temporal Convergence Trajectory (Secondary Analysis)

Objective

Determine how the DENV-ZIKV fold-change correlation changes across the four timepoints.

Input Files

- `01_processed_data/seurat_objects/sobj_processed.rds`

Analysis Procedure

```
# For each timepoint, compute fold changes and correlation
timepoints <- c("4h", "12h", "24h", "48h")
temporal_cors <- data.frame()

for (tp in timepoints) {
  sobj_tp <- subset(sobj, timepoint == tp)

  # Skip if insufficient cells
  if (sum(sobj_tp$condition == "DENV") < 30 |
      sum(sobj_tp$condition == "ZIKV") < 30 |
      sum(sobj_tp$condition == "Control") < 30) {
    cat("Skipping", tp, "- insufficient cells\n")
    next
  }

  fc_d <- FindMarkers(sobj_tp, ident.1 = "DENV", ident.2 = "Control",
                       group.by = "condition", logfc.threshold = 0,
                       min.pct = 0.1)
  fc_z <- FindMarkers(sobj_tp, ident.1 = "ZIKV", ident.2 = "Control",
                       group.by = "condition", logfc.threshold = 0,
                       min.pct = 0.1)

  fc_d$gene <- rownames(fc_d)
  fc_z$gene <- rownames(fc_z)

  m <- merge(fc_d[, c("gene", "avg_log2FC")],
             fc_z[, c("gene", "avg_log2FC")],
             by = "gene", suffixes = c("_DENV", "_ZIKV"))

  r <- cor(m$avg_log2FC_DENV, m$avg_log2FC_ZIKV, method = "pearson")

  temporal_cors <- rbind(temporal_cors, data.frame(
    timepoint = tp,
    pearson_r = r,
    n_genes = nrow(m)
  ))
}

# Plot temporal trajectory
ggplot(temporal_cors, aes(x = timepoint, y = pearson_r, group = 1)) +
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
  ylim(0, 1) +
  labs(x = "Time post-infection", y = "Pearson r (DENV vs ZIKV fold changes)",
       title = "Convergence trajectory") +
  theme_minimal()
ggsave("04_figures/supplementary/temporal_convergence.pdf", width = 6, height =
4)
```

```
write.csv(temporal_cors,
"03_results/phase3_shared_degs/temporal_convergence.csv",
          row.names = FALSE)
```

Interpretation

- r increasing over time: convergence emerges with viral replication (the expected pattern if driven by shared proviral exploitation)
- r high at all timepoints: convergence reflects shared innate sensing (less interesting for the exploitation story)
- r decreasing at 48h: late-stage divergence due to cytopathic effects or virus-specific processes

Things That Must Be Checked Before Proceeding to Phase 5

- ☐ Shared DEG lists saved (Step 4.1)
- ☐ Overlap significance documented (Step 4.2)
- ☐ Fold-change correlation computed and figure saved (Step 4.3)
- ☐ Gates G2 and G3 evaluated
- ☐ Temporal trajectory computed (Step 4.4)
- ☐ Decision to proceed documented with justification

PHASE 5 — HOST-FACTOR INTEGRATION

Step 5.1: Proviral Gene Enrichment Among Shared Upregulated DEGs (GATE G4)

Objective

Test whether genes upregulated by BOTH DENV and ZIKV are significantly enriched for experimentally validated proviral host factors.

Input Files

- `03_results/phase3_shared_degs/shared_DEGs_up.csv`
- `02_literature_resources/host_factors_proviral.txt`
- `02_literature_resources/host_factors_antiviral.txt`
- Background: all genes measured in GSE110496

Pre-Analysis Checks

- ☐ Shared DEG gene symbols match the format of host factor gene symbols
- ☐ Background gene list is from the same analysis (not a different dataset)
- ☐ Proviral and antiviral lists are non-overlapping

Analysis Procedure

```
# Load data
shared_up <- read.csv("03_results/phase3_shared_degs/shared_DEGs_up.csv")$gene
shared_down <- read.csv("03_results/phase3_shared_degs/shared_DEGs_down.csv")
$gene
proviral <- scan("02_literature_resources/host_factors_proviral.txt", what =
"character")
antiviral <- scan("02_literature_resources/host_factors_antiviral.txt", what =
"character")

# Background from the merged fold-change table
merged <-
read.csv("03_results/phase3_shared_degs/merged_foldchanges_annotated.csv")
background <- merged$gene
N <- length(background)

# Restrict to genes in the background
proviral_bg <- intersect(proviral, background)
antiviral_bg <- intersect(antiviral, background)

cat("Proviral genes in background:", length(proviral_bg), "/", length(proviral),
"\n")
cat("Antiviral genes in background:", length(antiviral_bg), "/",
length(antiviral), "\n")

# TEST 1: Proviral enrichment among shared UPREGULATED DEGs
n_shared_up <- length(shared_up)
n_proviral_in_up <- length(intersect(shared_up, proviral_bg))

fisher_prov_up <- fisher.test(matrix(c(
  n_proviral_in_up,
  n_shared_up - n_proviral_in_up,
  length(proviral_bg) - n_proviral_in_up,
  N - n_shared_up - length(proviral_bg) + n_proviral_in_up
), nrow = 2), alternative = "greater")

# TEST 2: Antiviral enrichment among shared DOWNREGULATED DEGs
n_shared_down <- length(shared_down)
n_antiviral_in_down <- length(intersect(shared_down, antiviral_bg))

fisher_anti_down <- fisher.test(matrix(c(
  n_antiviral_in_down,
  n_shared_down - n_antiviral_in_down,
  length(antiviral_bg) - n_antiviral_in_down,
  N - n_shared_down - length(antiviral_bg) + n_antiviral_in_down
), nrow = 2), alternative = "greater")

# TEST 3 (control): Proviral enrichment among shared DOWNREGULATED DEGs
# (should NOT be significant — proviral genes should not be downregulated)
n_proviral_in_down <- length(intersect(shared_down, proviral_bg))
fisher_prov_down <- fisher.test(matrix(c(
  n_proviral_in_down,
  n_shared_down - n_proviral_in_down,
  length(proviral_bg) - n_proviral_in_down,
```

```
  N - n_shared_down - length(proviral_bg) + n_proviral_in_down
), nrow = 2), alternative = "greater")

# Report
cat("\n=== HOST FACTOR ENRICHMENT RESULTS ===\n")
cat("Proviral in shared UP:", n_proviral_in_up, "/", n_shared_up,
    "| p =", fisher_prov_up$p.value,
    "| OR =", round(fisher_prov_up$estimate, 2), "\n")
cat("Antiviral in shared DOWN:", n_antiviral_in_down, "/", n_shared_down,
    "| p =", fisher_anti_down$p.value,
    "| OR =", round(fisher_anti_down$estimate, 2), "\n")
cat("Proviral in shared DOWN (control):", n_proviral_in_down, "/",
n_shared_down,
    "| p =", fisher_prov_down$p.value, "\n")

# GSEA with custom gene sets
library(clusterProfiler)

# Create ranked gene list (average FC across both viruses)
merged <-
read.csv("03_results/phase3_shared_degs/merged_foldchanges_annotated.csv")
avg_fc <- setNames((merged$avg_log2FC_DENV + merged$avg_log2FC_ZIKV) / 2,
                    merged$gene)
avg_fc <- sort(avg_fc, decreasing = TRUE)

# Custom gene sets in TERM2GENE format
term2gene <- data.frame(
  term = c(rep("Proviral", length(proviral_bg)),
           rep("Antiviral", length(antiviral_bg))),
  gene = c(proviral_bg, antiviral_bg)
)

gsea_hf <- GSEA(avg_fc, TERM2GENE = term2gene, pvalueCutoff = 1,
                 minGSSize = 5, maxGSSize = 500)

# Plot
gseaplot2(gsea_hf, geneSetID = "Proviral",
          title = "GSEA: Proviral host factors")
ggsave("04_figures/main/Figure3B_GSEA_proviral.pdf", width = 7, height = 5)

# Save all results
hf_results <- data.frame(
  Test = c("Proviral_in_SharedUp", "Antiviral_in_SharedDown",
           "Proviral_in_SharedDown_control"),
  Observed = c(n_proviral_in_up, n_antiviral_in_down, n_proviral_in_down),
  Total_Tested = c(n_shared_up, n_shared_down, n_shared_down),
  Fisher_p = c(fisher_prov_up$p.value, fisher_anti_down$p.value,
               fisher_prov_down$p.value),
  Odds_Ratio = c(fisher_prov_up$estimate, fisher_anti_down$estimate,
                  fisher_prov_down$estimate)
)
write.csv(hf_results, "03_results/phase4_host_factors/enrichment_results.csv",
          row.names = FALSE)

# List the specific proviral genes found in shared upregulated DEGs
proviral_hits <- intersect(shared_up, proviral_bg)
cat("\nProviral genes in shared upregulated DEGs:\n")
print(proviral_hits)
write.csv(data.frame(gene = proviral_hits),
          "03_results/phase4_host_factors/proviral_hits_in_shared_up.csv",
          row.names = FALSE)
```

Go / No-Go Decision (GATE G4)

STRONG SUPPORT if: Proviral enrichment $p < 0.01$, OR > 2, AND at least 5 proviral genes in shared upregulated DEGs. **MODERATE SUPPORT** if: $p < 0.05$ but fewer than 5 proviral genes. Report but note limited overlap. **WEAK SUPPORT** if: $p > 0.05$. The shared response is not preferentially proviral. Report the GSEA result as an alternative (GSEA may detect enrichment even when the binary overlap test does not). **NO SUPPORT** if: $p > 0.1$ and GSEA NES < 0 for proviral genes. The host factor layer does not connect to the transcriptomic data. Downweight the host factor story in the paper.

Output Files

- `03_results/phase4_host_factors/enrichment_results.csv`
- `03_results/phase4_host_factors/proviral_hits_in_shared_up.csv`
- `04_figures/main/Figure3B_GSEA_proviral.pdf`
- `04_figures/main/Figure3C_GSEA_antiviral.pdf`

PHASE 6 — miRNA INTEGRATION

Step 6.1: Three-Tier miRNA Target Enrichment (GATE G5)

Objective

Test whether targets of DENV-downregulated miRNAs are enriched among shared upregulated DEGs. This is the mechanistic test: if miRNAs are suppressed, their targets should be de-repressed (upregulated).

Input Files

- `03_results/phase3_shared_degs/shared_DEGs_up.csv`
- `02_literature_resources/mirna_targets_55_highconf.txt` (PRIMARY)
- `02_literature_resources/mirna_targets_high_conf.txt` (all 91 — SENSITIVITY)
- `02_literature_resources/mirna_targets_36_highconf.txt` (SPECIFICITY CONTROL)
- Background gene list

Pre-Analysis Checks

- ☐ Target lists are non-empty
- ☐ Target gene symbols match DEG gene symbols (same format)

- ☐ The 55-miRNA targets and 36-miRNA targets are subsets of the 91-miRNA targets

Analysis Procedure

```
shared_up <- read.csv("03_results/phase3_shared_degs/shared_DEGs_up.csv")$gene
background <- merged$gene
N <- length(background)

# Load three target sets
targets_55 <- scan("02_literature_resources/mirna_targets_55_highconf.txt",
                    what = "character")
targets_91 <- scan("02_literature_resources/mirna_targets_high_conf.txt",
                    what = "character")
targets_36 <- scan("02_literature_resources/mirna_targets_36_highconf.txt",
                    what = "character")

# Restrict to background
t55_bg <- intersect(targets_55, background)
t91_bg <- intersect(targets_91, background)
t36_bg <- intersect(targets_36, background)

# Function to run enrichment test
run_enrichment <- function(targets_in_bg, shared_genes, bg_size, label) {
  n_shared <- length(shared_genes)
  n_target <- length(targets_in_bg)
  n_overlap <- length(intersect(shared_genes, targets_in_bg))
  expected <- (n_target * n_shared) / bg_size

  ft <- fisher.test(matrix(c(
    n_overlap,
    n_shared - n_overlap,
    n_target - n_overlap,
    bg_size - n_shared - n_target + n_overlap
  ), nrow = 2), alternative = "greater")

  data.frame(
    Set = label,
    N_targets_in_bg = n_target,
    N_shared_up = n_shared,
    Overlap = n_overlap,
    Expected = round(expected, 1),
    Fold_Enrichment = round(n_overlap / expected, 2),
    Fisher_p = ft$p.value,
    Odds_Ratio = round(ft$estimate, 2)
  )
}

# PRIMARY: 55-miRNA set
result_55 <- run_enrichment(t55_bg, shared_up, N, "55_miRNA_primary")

# SENSITIVITY: 91-miRNA set
result_91 <- run_enrichment(t91_bg, shared_up, N, "91_miRNA_sensitivity")

# SPECIFICITY CONTROL: 36-miRNA set
result_36 <- run_enrichment(t36_bg, shared_up, N, "36_miRNA_specificity")

# Combine and display
mirna_results <- rbind(result_55, result_91, result_36)
print(mirna_results)
write.csv(mirna_results, "03_results/phase5_mirna/three_tier_enrichment.csv",
          row.names = FALSE)

# INTERPRETATION GUIDE:
```

```
# If 55 significant AND 36 NOT significant: specificity confirmed
# If 55 significant AND 91 significant: robust finding
# If 55 significant AND 36 also significant: no specificity (generic miRNA effect)
# If 55 NOT significant: miRNA layer does not connect to transcriptomics
```

Quality Control Checks

- ☐ The number of target genes in the background is reasonable (55-set: expect 1500-3000; 36-set: expect 1000-2000)
- ☐ The overlap count is not zero for the primary (55) analysis
- ☐ The fold enrichment makes biological sense (expect 1.3-3.0 range)

Go / No-Go Decision (GATE G5)

STRONG MECHANISTIC EVIDENCE if: 55-miRNA $p < 0.01$ AND 36-miRNA $p > 0.1$. The enrichment is specific to cross-flavivirus miRNAs. Feature miRNAs prominently in the paper. **MODERATE EVIDENCE** if: 55-miRNA $p < 0.05$ regardless of 36 result. Include miRNA layer but without the specificity argument. **WEAK EVIDENCE** if: 55-miRNA p = 0.05-0.1. Mention as "trend" but do not build the narrative around it. **NO EVIDENCE** if: 55-miRNA $p > 0.1$. The miRNA layer does not explain the shared response. Dramatically reduce the miRNA emphasis in the paper. Report as a negative finding: "miRNA-mediated regulation was tested but not supported at the statistical level."

Step 6.2: Multi-Layer Intersection

Objective

Identify genes that satisfy all available evidence criteria simultaneously.

Input Files

- `03_results/phase3_shared_degs/shared_DEGs_up.csv`
- `02_literature_resources/host_factors_proviral.txt`
- `02_literature_resources/mirna_targets_55_highconf.txt`
- Background gene list

Analysis Procedure

```
library(SuperExactTest)
library(UpSetR)

# Only perform if BOTH G4 and G5 passed
# (no point computing a three-way intersection if one layer is not significant)

shared_up <- read.csv("03_results/phase3_shared_degs/shared_DEGs_up.csv")$gene
proviral_bg <- intersect(proviral, background)
targets_55_bg <- intersect(targets_55, background)
```

```r
# Three-way intersection
three_way <- Reduce(intersect, list(shared_up, proviral_bg, targets_55_bg))

cat("Three-way intersection genes:", length(three_way), "\n")
print(three_way)

# SuperExactTest for statistical significance
input_list <- list(
  `Shared UP DEGs` = shared_up,
  `Proviral factors` = proviral_bg,
  `miRNA-55 targets` = targets_55_bg
)

super_result <- supertest(input_list, n = N)
summary(super_result)

pdf("04_figures/main/Figure3A_SuperExactTest.pdf", width = 7, height = 6)
plot(super_result, sort.by = "size")
dev.off()

# Annotate three-way genes
if (length(three_way) > 0) {
  three_way_annot <- data.frame(
    gene = three_way,
    DENV_log2FC = merged$avg_log2FC_DENV[match(three_way, merged$gene)],
    ZIKV_log2FC = merged$avg_log2FC_ZIKV[match(three_way, merged$gene)],
    proviral_mechanism = host_factors$mechanism[
      match(three_way, host_factors$gene_symbol)],
    pathway = host_factors$pathway[match(three_way, host_factors$gene_symbol)]
  )
  write.csv(three_way_annot,
            "03_results/phase5_mirna/three_way_intersection_annotated.csv",
            row.names = FALSE)
}

# UpSet plot of all intersections
all_genes_bg <- unique(c(shared_up, proviral_bg, targets_55_bg))
upset_df <- data.frame(
  gene = all_genes_bg,
  Shared_UP = as.integer(all_genes_bg %in% shared_up),
  Proviral = as.integer(all_genes_bg %in% proviral_bg),
  miRNA_targets = as.integer(all_genes_bg %in% targets_55_bg)
)
pdf("04_figures/main/Figure3A_UpSet_ThreeWay.pdf", width = 8, height = 5)
upset(upset_df[, -1], order.by = "freq")
dev.off()
```

Output Files

- `03_results/phase5_mirna/three_way_intersection_annotated.csv`
- `04_figures/main/Figure3A_SuperExactTest.pdf`
- `04_figures/main/Figure3A_UpSet_ThreeWay.pdf`

Interpretation

The three-way intersection genes are the strongest candidates in the study. They are:

1. Transcriptomically shared between DENV and ZIKV (Layer 2)
2. Independently classified as proviral (Layer 3)

3. Targets of DENV-downregulated miRNAs with ZIKV genome binding (Layer 1)

These genes become Table 3 in the manuscript and are highlighted by name in the scatter plot and network figures.

Things That Must Be Checked Before Proceeding to Phase 7 (Pathway Analysis)

- ☐ All enrichment tests completed (Steps 5.1 and 6.1)
- ☐ Gates G4 and G5 evaluated and documented
- ☐ Three-way intersection computed and annotated
- ☐ All results saved in the expected directories

STANDARD OPERATING PROCEDURE — Part 2

Phases 7-12: Pathway Analysis through Final Interpretation

PHASE 7 — PATHWAY ANALYSIS

Step 7.1: Pathway Enrichment of Shared DEGs

Objective

Identify which biological pathways are represented by genes shared between DENV and ZIKV responses. This provides functional interpretation of the convergence finding.

Input Files

- `03_results/phase3_shared_degs/shared_DEGs_up.csv`
- `03_results/phase3_shared_degs/shared_DEGs_down.csv`
- `03_results/phase3_shared_degs/shared_DEGs_all.csv`

Pre-Analysis Checks

- ☐ clusterProfiler, ReactomePA, and org.Hs.eg.db installed
- ☐ At least 30 shared DEGs exist (fewer makes enrichment analysis underpowered)
- ☐ Gene symbols are HGNC approved (required for ID conversion)

Analysis Procedure

```
library(clusterProfiler)
library(org.Hs.eg.db)
```

```
library(ReactomePA)
library(enrichplot)

# Load shared DEGs
shared_all <- read.csv("03_results/phase3_shared_degs/shared_DEGs_all.csv")
shared_up <- read.csv("03_results/phase3_shared_degs/shared_DEGs_up.csv")$gene
shared_down <- read.csv("03_results/phase3_shared_degs/shared_DEGs_down.csv")
$gene

# Convert to Entrez IDs
convert <- bitr(shared_all$gene, fromType = "SYMBOL",
                toType = "ENTREZID", OrgDb = org.Hs.eg.db)
shared_entrez <- convert$ENTREZID

up_convert <- bitr(shared_up, fromType = "SYMBOL",
                    toType = "ENTREZID", OrgDb = org.Hs.eg.db)
down_convert <- bitr(shared_down, fromType = "SYMBOL",
                      toType = "ENTREZID", OrgDb = org.Hs.eg.db)

cat("Gene mapping success rate:",
    round(100 * nrow(convert) / nrow(shared_all), 1), "%\n")
# If < 80%: check for alias symbols

# === KEGG Enrichment ===
kegg_all <- enrichKEGG(gene = shared_entrez, organism = "hsa",
                        pvalueCutoff = 0.05, pAdjustMethod = "BH",
                        minGSSize = 10, maxGSSize = 500)
kegg_up <- enrichKEGG(gene = up_convert$ENTREZID, organism = "hsa",
                       pvalueCutoff = 0.05)
kegg_down <- enrichKEGG(gene = down_convert$ENTREZID, organism = "hsa",
                         pvalueCutoff = 0.05)

# === GO Biological Process ===
go_bp <- enrichGO(gene = shared_entrez, OrgDb = org.Hs.eg.db,
                   ont = "BP", pAdjustMethod = "BH",
                   pvalueCutoff = 0.05, readable = TRUE,
                   minGSSize = 10, maxGSSize = 500)

# Remove redundant GO terms
go_bp_simplified <- simplify(go_bp, cutoff = 0.7, by = "p.adjust",
                              select_fun = min)

# === Reactome ===
reactome <- enrichPathway(gene = shared_entrez, organism = "human",
                           pvalueCutoff = 0.05, readable = TRUE,
                           minGSSize = 10, maxGSSize = 500)

# === Visualization ===
if (nrow(as.data.frame(kegg_all)) > 0) {
  dotplot(kegg_all, showCategory = 15, title = "KEGG: Shared DENV-ZIKV DEGs")
  ggsave("04_figures/main/Figure4A_KEGG_shared.pdf", width = 8, height = 6)
}

if (nrow(as.data.frame(go_bp_simplified)) > 0) {
  dotplot(go_bp_simplified, showCategory = 15, title = "GO:BP: Shared DEGs")
  ggsave("04_figures/supplementary/GO_BP_shared.pdf", width = 8, height = 6)
}

if (nrow(as.data.frame(reactome)) > 0) {
  dotplot(reactome, showCategory = 15, title = "Reactome: Shared DEGs")
  ggsave("04_figures/supplementary/Reactome_shared.pdf", width = 8, height = 6)
}

# === Save all results ===
```

```
write.csv(as.data.frame(kegg_all),
          "03_results/phase6_pathways/KEGG_shared_all.csv", row.names = FALSE)
write.csv(as.data.frame(kegg_up),
          "03_results/phase6_pathways/KEGG_shared_up.csv", row.names = FALSE)
write.csv(as.data.frame(kegg_down),
          "03_results/phase6_pathways/KEGG_shared_down.csv", row.names = FALSE)
write.csv(as.data.frame(go_bp_simplified),
          "03_results/phase6_pathways/GO_BP_shared.csv", row.names = FALSE)
write.csv(as.data.frame(reactome),
          "03_results/phase6_pathways/Reactome_shared.csv", row.names = FALSE)
```

Quality Control Checks

- ☐ Gene ID conversion success rate > 80%
- ☐ At least 3 significantly enriched KEGG pathways (padj < 0.05)
- ☐ Enriched pathways include biologically expected terms for flavivirus infection. Expected: interferon signaling, TNF signaling, apoptosis, NF-kB, protein processing in ER, autophagy, JAK-STAT. Presence of at least 2 of these validates that the shared DEGs are biologically coherent.
- ☐ No obvious artifacts: "Ribosome" or "Spliceosome" alone (these are generic stress indicators, not infection-specific)
- ☐ GO term simplification reduced redundancy (fewer terms than the raw result)

COMMON MISTAKE: Running pathway enrichment on shared DEGs without specifying a universe/background. clusterProfiler uses the full genome as background by default, which is appropriate for this analysis. Do NOT set universe to "only measured genes" unless you have a specific reason.

Output Files

- `03_results/phase6_pathways/KEGG_shared_all.csv`
- `03_results/phase6_pathways/KEGG_shared_up.csv`
- `03_results/phase6_pathways/KEGG_shared_down.csv`
- `03_results/phase6_pathways/GO_BP_shared.csv`
- `03_results/phase6_pathways/Reactome_shared.csv`
- `04_figures/main/Figure4A_KEGG_shared.pdf`

Things That Must Be Checked Before Proceeding

- ☐ Pathway results reviewed for biological coherence
- ☐ Expected flavivirus-relevant pathways identified
- ☐ Results saved

PHASE 8 — EXTERNAL VALIDATION

Step 8.1: DEA on GSE118305 (ZIKV Macrophages)

Objective

Perform differential expression analysis on the ZIKV macrophage dataset to generate an independent ZIKV DEG list.

Input Files

- `00_raw_data/GSE118305/expression_matrix_rnaseq.csv`
- `00_raw_data/GSE118305/sample_metadata.csv`

Pre-Analysis Checks

- ☐ Confirmed which samples are ZIKV+ (infected) vs. Mock (control)
- ☐ At least 3 biological replicates per group
- ☐ Data format identified: raw counts (use DESeq2) or FPKM (use limma-voom)

Analysis Procedure

```
# If RAW COUNTS available:
library(DESeq2)

counts <- read.csv("00_raw_data/GSE118305/expression_matrix_rnaseq.csv",
                    row.names = 1)
meta <- read.csv("00_raw_data/GSE118305/sample_metadata.csv", row.names = 1)

# Select ZIKV+ vs Mock at 24h timepoint
selected <- meta[meta$condition %in% c("ZIKV_positive", "Mock") &
                  meta$timepoint == "24h", ]
counts_selected <- counts[, rownames(selected)]

dds <- DESeqDataSetFromMatrix(
  countData = counts_selected,
  colData = selected,
  design = ~ condition
)
dds <- DESeq(dds)
res <- results(dds, contrast = c("condition", "ZIKV_positive", "Mock"))
res_df <- as.data.frame(res)
res_df$gene <- rownames(res_df)

write.csv(res_df,
          "01_processed_data/deg_tables/DEGs_ZIKV_macrophages_GSE118305.csv",
          row.names = FALSE)

# If FPKM only:
library(limma)
# Use limma-voom workflow with log2-transformed FPKM values
```

Output Files

- `01_processed_data/deg_tables/DEGs_ZIKV_macrophages_GSE118305.csv`

Step 8.2: DEA on GSE94892 (DENV Patient PBMCs)

Objective

Perform DEA on the DENV patient PBMC dataset.

Input Files

- `00_raw_data/GSE94892/expression_matrix.csv`
- `00_raw_data/GSE94892/patient_metadata.csv`

Analysis Procedure

```
library(DESeq2)

counts <- read.csv("00_raw_data/GSE94892/expression_matrix.csv", row.names = 1)
meta <- read.csv("00_raw_data/GSE94892/patient_metadata.csv", row.names = 1)

# Compare all DENV patients vs controls
dds <- DESeqDataSetFromMatrix(
  countData = counts,
  colData = meta,
  design = ~ condition  # "DENV" vs "Control"
)
dds <- DESeq(dds)
res <- results(dds, contrast = c("condition", "DENV", "Control"))
res_df <- as.data.frame(res)
res_df$gene <- rownames(res_df)

write.csv(res_df,
          "01_processed_data/deg_tables/DEGs_DENV_PBMCs_GSE94892.csv",
          row.names = FALSE)
```

Output Files

- `01_processed_data/deg_tables/DEGs_DENV_PBMCs_GSE94892.csv`

Step 8.3: Replication Testing (GATE G6)

Objective

Test whether the shared DEGs from the discovery dataset (GSE110496) are also perturbed in independent immune cell datasets.

Input Files

- `03_results/phase3_shared_degs/shared_DEGs_up.csv`

- `01_processed_data/deg_tables/DEGs_ZIKV_macrophages_GSE118305.csv`
- `01_processed_data/deg_tables/DEGs_DENV_PBMCs_GSE94892.csv`

Pre-Analysis Checks

- ☐ Gene symbols are consistent across all three datasets
- ☐ Validation DEG tables have padj and log2FC columns

Analysis Procedure

```
# Load discovery shared DEGs
discovery_up <- read.csv("03_results/phase3_shared_degs/shared_DEGs_up.csv")
$gene
discovery_down <- read.csv("03_results/phase3_shared_degs/shared_DEGs_down.csv")
$gene

# Load validation DEGs
val_zikv <-
read.csv("01_processed_data/deg_tables/DEGs_ZIKV_macrophages_GSE118305.csv")
val_denv <-
read.csv("01_processed_data/deg_tables/DEGs_DENV_PBMCs_GSE94892.csv")

# Significant DEGs in validation datasets
zikv_val_up <- val_zikv$gene[val_zikv$log2FoldChange > 0 &
                              !is.na(val_zikv$padj) & val_zikv$padj < 0.05]
denv_val_up <- val_denv$gene[val_denv$log2FoldChange > 0 &
                              !is.na(val_denv$padj) & val_denv$padj < 0.05]

# Gene-level replication
rep_zikv <- intersect(discovery_up, zikv_val_up)
rep_denv <- intersect(discovery_up, denv_val_up)
rep_both <- intersect(rep_zikv, rep_denv)

cat("Discovery shared up:", length(discovery_up), "\n")
cat("Replicated in ZIKV macrophages:", length(rep_zikv),
    "(", round(100*length(rep_zikv)/length(discovery_up), 1), "%)\n")
cat("Replicated in DENV PBMCs:", length(rep_denv),
    "(", round(100*length(rep_denv)/length(discovery_up), 1), "%)\n")
cat("Replicated in both:", length(rep_both), "\n")

# Statistical test: enrichment of discovery genes in validation DEGs
# ZIKV validation
bg_zikv <- intersect(discovery_up, val_zikv$gene)  # genes measured in both
n_bg <- length(bg_zikv)
n_val_up <- length(intersect(zikv_val_up, bg_zikv))
n_disc <- length(intersect(discovery_up, bg_zikv))
n_overlap <- length(rep_zikv)

fisher_val_zikv <- fisher.test(matrix(c(
  n_overlap, n_disc - n_overlap,
  n_val_up - n_overlap, n_bg - n_disc - n_val_up + n_overlap
), nrow = 2), alternative = "greater")

cat("Fisher p (ZIKV validation):", fisher_val_zikv$p.value, "\n")

# Repeat for DENV validation
# (same structure)

# Pathway-level replication
```

```
# Run pathway enrichment on validation DEGs
# Compare enriched pathways with Phase 7 results
library(clusterProfiler)
library(org.Hs.eg.db)

# ZIKV validation pathways
zikv_val_entrez <- bitr(zikv_val_up, fromType = "SYMBOL",
                         toType = "ENTREZID", OrgDb = org.Hs.eg.db)$ENTREZID
kegg_zikv_val <- enrichKEGG(gene = zikv_val_entrez, organism = "hsa",
                              pvalueCutoff = 0.05)

# Compare pathway overlap with discovery
discovery_pathways <- read.csv("03_results/phase6_pathways/KEGG_shared_all.csv")
$ID
validation_pathways <- as.data.frame(kegg_zikv_val)$ID

shared_pathways <- intersect(discovery_pathways, validation_pathways)
cat("Shared KEGG pathways:", length(shared_pathways), "\n")
cat("Pathway names:", "\n")
print(as.data.frame(kegg_zikv_val)[as.data.frame(kegg_zikv_val)$ID %in%
  shared_pathways, c("ID", "Description")])

# Save results
val_results <- data.frame(
  Metric = c("Discovery_shared_up", "Replicated_ZIKV", "Replicated_DENV",
             "Replicated_both", "Fisher_p_ZIKV", "Fisher_p_DENV",
             "Shared_KEGG_pathways"),
  Value = c(length(discovery_up), length(rep_zikv), length(rep_denv),
            length(rep_both), fisher_val_zikv$p.value, NA,
            length(shared_pathways))
)
write.csv(val_results,
          "03_results/phase7_validation/replication_results.csv",
          row.names = FALSE)
```

Go / No-Go Decision (GATE G6)

STRONG REPLICATION if: Fisher p < 0.05 for at least one validation dataset AND >= 3 shared KEGG pathways. **MODERATE REPLICATION** if: Gene-level replication rate > 10% in at least one dataset OR >= 2 shared pathways. **WEAK REPLICATION** if: Gene-level replication < 10% AND < 2 shared pathways. Note cell-type limitations in the paper. The finding is specific to Huh7 cells and may not generalize. **NO REPLICATION** if: Replication rate < 5% AND zero shared pathways. Limit all claims to the GSE110496 system.

Output Files

- `03_results/phase7_validation/replication_results.csv`
- `03_results/phase7_validation/replicated_genes_ZIKV.csv`
- `03_results/phase7_validation/replicated_genes_DENV.csv`
- `03_results/phase7_validation/pathway_comparison.csv`

Step 8.4: Neural Extension Analysis (GSE78711)

Objective

Test whether the shared DENV-ZIKV host response identified in Huh7 cells extends to neural progenitor cells, the cell type most relevant to ZIKV-induced microcephaly. This is a one-sided test: only ZIKV infects neural progenitor cells (DENV does not), so this step asks whether the shared gene set is perturbed in the neural context, not whether DENV-ZIKV convergence exists in neural cells.

Input Files

- `00_raw_data/GSE78711/expression_matrix.csv`
- `00_raw_data/GSE78711/sample_metadata.csv`
- `03_results/phase3_shared_degs/shared_DEGs_up.csv`
- `03_results/phase3_shared_degs/shared_DEGs_down.csv`

Pre-Analysis Checks

- ☐ GSE78711 expression data successfully downloaded (Step 1.4)
- ☐ At least 2 replicates per condition (ZIKV vs. mock)
- ☐ Gene identifiers match the format used in the discovery analysis (convert if needed)
- ☐ Phase 4 shared DEGs have already been identified

Analysis Procedure

```
# DEA on GSE78711
# If raw counts: use DESeq2
# If FPKM only: use limma

# Assuming counts are available:
library(DESeq2)

counts <- read.csv("00_raw_data/GSE78711/expression_matrix.csv", row.names = 1)
meta <- read.csv("00_raw_data/GSE78711/sample_metadata.csv", row.names = 1)

# Ensure condition column exists with "ZIKV" and "Mock" labels
dds <- DESeqDataSetFromMatrix(
  countData = counts,
  colData = meta,
  design = ~ condition
)
dds <- DESeq(dds)
res <- results(dds, contrast = c("condition", "ZIKV", "Mock"))
res_df <- as.data.frame(res)
res_df$gene <- rownames(res_df)

write.csv(res_df,
          "01_processed_data/deg_tables/DEGs_ZIKV_NPCs_GSE78711.csv",
          row.names = FALSE)

# Identify upregulated genes in ZIKV-infected NPCs
npc_up <- res_df$gene[!is.na(res_df$padj) &
                       res_df$padj < 0.05 &
```

```
                       res_df$log2FoldChange >= 1]
npc_down <- res_df$gene[!is.na(res_df$padj) &
                         res_df$padj < 0.05 &
                         res_df$log2FoldChange <= -1]

cat("ZIKV NPC DEGs up:", length(npc_up), "\n")
cat("ZIKV NPC DEGs down:", length(npc_down), "\n")

# Test: Are the discovery shared upregulated DEGs also upregulated in NPCs?
discovery_up <- read.csv("03_results/phase3_shared_degs/shared_DEGs_up.csv")
$gene

# Restrict to genes measured in both datasets
bg_npc <- intersect(discovery_up, res_df$gene)
rep_in_npc <- intersect(discovery_up, npc_up)

cat("Discovery shared up genes measured in NPC data:", length(bg_npc), "\n")
cat("Also upregulated in NPCs:", length(rep_in_npc),
    "(", round(100 * length(rep_in_npc) / length(bg_npc), 1), "%)\n")

# Fisher's exact test
all_genes_npc <- res_df$gene
N_npc <- length(all_genes_npc)
n_npc_up <- length(npc_up)
n_disc_in_npc <- length(intersect(discovery_up, all_genes_npc))
n_overlap_npc <- length(rep_in_npc)

fisher_npc <- fisher.test(matrix(c(
  n_overlap_npc,
  n_disc_in_npc - n_overlap_npc,
  n_npc_up - n_overlap_npc,
  N_npc - n_disc_in_npc - n_npc_up + n_overlap_npc
), nrow = 2), alternative = "greater")

cat("Fisher p-value (neural extension):", fisher_npc$p.value, "\n")
cat("Odds ratio:", round(fisher_npc$estimate, 2), "\n")

# Also test proviral gene enrichment in NPC DEGs
proviral <- scan("02_literature_resources/host_factors_proviral.txt", what =
"character")
proviral_in_npc_up <- intersect(npc_up, proviral)
cat("Proviral genes upregulated in ZIKV NPCs:", length(proviral_in_npc_up),
"\n")

# Pathway comparison
library(clusterProfiler)
library(org.Hs.eg.db)

npc_up_entrez <- bitr(npc_up, fromType = "SYMBOL",
                       toType = "ENTREZID", OrgDb = org.Hs.eg.db)$ENTREZID
kegg_npc <- enrichKEGG(gene = npc_up_entrez, organism = "hsa", pvalueCutoff =
0.05)

# Compare with discovery shared DEG pathways
discovery_pathways <- read.csv("03_results/phase6_pathways/KEGG_shared_all.csv")
$ID
npc_pathways <- as.data.frame(kegg_npc)$ID
shared_pathways_npc <- intersect(discovery_pathways, npc_pathways)
cat("Shared pathways with discovery:", length(shared_pathways_npc), "\n")

# Save results
npc_results <- data.frame(
  Metric = c("NPC_DEGs_up", "NPC_DEGs_down",
             "Discovery_shared_up_in_NPC_bg", "Replicated_in_NPC",
```

```
             "Replication_rate_pct", "Fisher_p", "Odds_ratio",
             "Proviral_in_NPC_up", "Shared_KEGG_pathways"),
  Value = c(length(npc_up), length(npc_down),
            length(bg_npc), length(rep_in_npc),
            round(100 * length(rep_in_npc) / max(length(bg_npc), 1), 1),
            fisher_npc$p.value, round(fisher_npc$estimate, 2),
            length(proviral_in_npc_up), length(shared_pathways_npc))
)
write.csv(npc_results,
          "03_results/phase7_validation/neural_extension_results.csv",
          row.names = FALSE)
write.csv(data.frame(gene = rep_in_npc),
          "03_results/phase7_validation/replicated_genes_NPCs.csv",
          row.names = FALSE)
```

Quality Control Checks

- ☐ DEA on GSE78711 produced a reasonable number of DEGs (expect 500-3000 given ZIKV actively replicates in NPCs)
- ☐ The overlap is tested against the correct background (genes measured in BOTH discovery and NPC datasets)
- ☐ Gene symbol formats are consistent between datasets

Interpretation

This step answers: does the conserved flavivirus host response extend to the neural tissue where ZIKV causes its most devastating pathology?

If Fisher p < 0.05: The shared DENV-ZIKV gene set is also perturbed during ZIKV infection of neural progenitor cells. This demonstrates that the conserved host exploitation program operates across hepatoma cells, immune cells, and neural cells, suggesting it is a fundamental feature of flavivirus biology rather than a cell-type-specific artifact. This is a strong result for the paper.

If Fisher p > 0.05 but pathway overlap exists: The convergence extends to neural cells at the pathway level but not at the individual gene level. This is expected: different cell types express different gene repertoires, but the same pathways can be perturbed through different gene members. Report pathway-level conservation.

If no gene or pathway overlap: The shared response does not extend to neural cells. This is an informative negative result: it may mean the convergent proviral program is specific to cells where both viruses replicate (immune cells and hepatocytes) and does not govern neural tropism. Report honestly.

Important Caveat for the Manuscript

GSE78711 uses ZIKV strain MR766 (African lineage, isolated 1947), while the 2015-2016 epidemic was caused by Asian lineage strains (PRVABC59 and related). Transcriptomic differences between lineages have been documented. The manuscript should note: "Neural extension analysis was performed using GSE78711, which employed the MR766 strain. Validation with Asian lineage ZIKV strains in neural progenitor cells would strengthen these findings."

Output Files

- `01_processed_data/deg_tables/DEGs_ZIKV_NPCs_GSE78711.csv`
- `03_results/phase7_validation/neural_extension_results.csv`
- `03_results/phase7_validation/replicated_genes_NPCs.csv`

Things That Must Be Checked Before Proceeding to Phase 9

- ☐ All six gates evaluated and documented
- ☐ Neural extension results documented
- ☐ Results summary table created showing pass/fail for each gate
- ☐ Decision to proceed to network analysis justified by gate results

---

PHASE 9 — NETWORK ANALYSIS

PREREQUISITE: Proceed ONLY if Gates G2, G3, and at least one of G4/G5 passed. If all failed except G2/G3, the network analysis will visualize the convergence finding but will lack the functional annotation that makes networks biologically interpretable.

Step 9.1: PPI Network Construction

Objective

Build a protein-protein interaction network of the shared DEGs and overlay functional annotations.

Input Files

- `03_results/phase3_shared_degs/shared_DEGs_all.csv`
- `03_results/phase3_shared_degs/merged_foldchanges_annotated.csv`
- `03_results/phase4_host_factors/proviral_hits_in_shared_up.csv`
- `03_results/phase5_mirna/three_way_intersection_annotated.csv` (if exists)

Analysis Procedure

```
library(STRINGdb)

# Initialize STRING
string_db <- STRINGdb$new(version = "12.0", species = 9606,
                          score_threshold = 400)

# Map genes
shared_genes <- read.csv("03_results/phase3_shared_degs/shared_DEGs_all.csv")
mapped <- string_db$map(shared_genes, "gene", removeUnmappedRows = TRUE)

cat("Genes mapped to STRING:", nrow(mapped), "/", nrow(shared_genes), "\n")
```

```
# Get interactions
if (nrow(mapped) > 1) {
  interactions <- string_db$get_interactions(mapped$STRING_id)
  cat("PPI interactions found:", nrow(interactions), "\n")

  # Export edge list for Cytoscape
  # Convert STRING IDs back to gene symbols
  id_map <- setNames(mapped$gene, mapped$STRING_id)
  interactions$gene1 <- id_map[interactions$from]
  interactions$gene2 <- id_map[interactions$to]

  edges <- interactions[, c("gene1", "gene2", "combined_score")]
  write.csv(edges,
            "03_results/phase8_network/network_edges.csv",
            row.names = FALSE)
}

# Create node attribute table for Cytoscape
merged_data <-
read.csv("03_results/phase3_shared_degs/merged_foldchanges_annotated.csv")
proviral <- scan("02_literature_resources/host_factors_proviral.txt", what =
"character")
antiviral <- scan("02_literature_resources/host_factors_antiviral.txt", what =
"character")
mirna_targets <- scan("02_literature_resources/mirna_targets_55_highconf.txt",
                       what = "character")

three_way_genes <- tryCatch(
  read.csv("03_results/phase5_mirna/three_way_intersection_annotated.csv")$gene,
  error = function(e) character(0)
)

node_attrs <- data.frame(
  gene = shared_genes$gene,
  direction = shared_genes$direction,
  DENV_log2FC = merged_data$avg_log2FC_DENV[match(shared_genes$gene,
merged_data$gene)],
  ZIKV_log2FC = merged_data$avg_log2FC_ZIKV[match(shared_genes$gene,
merged_data$gene)],
  is_proviral = shared_genes$gene %in% proviral,
  is_antiviral = shared_genes$gene %in% antiviral,
  is_mirna_target = shared_genes$gene %in% mirna_targets,
  is_three_way = shared_genes$gene %in% three_way_genes,
  avg_FC = merged_data$avg_log2FC_DENV[match(shared_genes$gene,
merged_data$gene)] +
           merged_data$avg_log2FC_ZIKV[match(shared_genes$gene,
merged_data$gene)]
)
node_attrs$avg_FC <- node_attrs$avg_FC / 2

write.csv(node_attrs,
          "03_results/phase8_network/network_node_attributes.csv",
          row.names = FALSE)
```

Cytoscape Procedure (Manual Steps)

1. Open Cytoscape
2. File > Import > Network from File > select `network_edges.csv`
    - Source: gene1, Target: gene2, Edge attribute: combined_score

3. File > Import > Table from File > select `network_node_attributes.csv`
    - Import to: Node Table, Key column: gene
4. Apply layout: Layout > Prefuse Force Directed > select "combined_score" as weight
5. Visual mapping (Style panel):
    - Node Fill Color: Map "is_proviral" (TRUE = #D73027 red, FALSE = check next)
    - If not proviral, map "is_antiviral" (TRUE = #4575B4 blue, FALSE = #BDBDBD grey)
    - Node Size: Map "avg_FC" to size (continuous, larger = more upregulated)
    - Node Border Width: Map "is_three_way" (TRUE = 5px, FALSE = 1px)
    - Node Shape: Map "is_mirna_target" (TRUE = Diamond, FALSE = Ellipse)
    - Edge Width: Map "combined_score" to width (continuous)
    - Edge Color: #CCCCCC
6. Run NetworkAnalyzer: Tools > Analyze Network
    - Record: number of nodes, edges, clustering coefficient, average degree
7. Install and run cytoHubba: Apps > cytoHubba
    - Method: MCC (Maximal Clique Centrality)
    - Export top 10 hub genes
8. Install and run MCODE: Apps > MCODE
    - Parameters: Node score cutoff = 0.2, K-core = 2, Degree cutoff = 2
    - Export modules
9. Export figure: File > Export as Image > PDF, 300 DPI

What NOT to Overinterpret

- Hub gene identification is topology-dependent. Different centrality metrics (degree, betweenness, MCC) may give different rankings. Report results from MCC as primary but note discrepancies.
- MCODE modules depend on parameter settings. The default parameters may not produce biologically meaningful modules. Check module membership against known pathways before interpreting.
- The network is built from STRING's database, which reflects published literature. Well-studied genes will have more interactions regardless of their importance in this specific system.

Output Files

- `03_results/phase8_network/network_edges.csv`
- `03_results/phase8_network/network_node_attributes.csv`
- `03_results/phase8_network/hub_genes_MCC_top10.csv`

- `03_results/phase8_network/mcode_modules.csv`
- `03_results/phase8_network/network_statistics.csv`
- `04_figures/main/Figure5_Network.pdf`

PHASE 10 — OPTIONAL ADVANCED ANALYSES

These analyses are OPTIONAL. Perform only after Phases 3-9 are complete and the manuscript draft is underway. Skip if time is limited. None of these are required for publication.

Step 10.1: Viral Load Correlation Analysis

When to Perform

Only if: (a) GSE110496 contains per-cell viral RNA quantification, (b) Phase 3-5 showed strong convergence, (c) you want a secondary figure adding mechanistic depth.

What It Adds

Identifies genes whose expression correlates with intracellular viral load within infected cells. Genes positively correlated with viral load are candidate proviral factors (their expression increases as the virus replicates more). This is more informative than binary infected/control comparison because it captures dose-response relationships.

When to Skip

If viral read counts are not available in GSE110496 metadata. If Phase 3 convergence was weak. If the paper is already sufficiently strong without this analysis.

Step 10.2: WGCNA on GSE118305

When to Perform

Only if: (a) All gates G1-G6 passed, (b) GSE118305 has >= 30 samples in the ZIKV and control groups combined, (c) you want to test whether co-expression modules (not just individual genes) are conserved.

What It Adds

One additional sentence: "Not only are individual genes shared, but their co-expression architecture is preserved across the two viruses." This is nice but not essential.

When to Skip

If any gate failed. If sample size is inadequate for WGCNA (< 20 samples per condition). If the paper is already strong without module analysis. If a reviewer requests it during revision, it can be added then.

Step 10.3: Pseudotime Analysis

When to Perform

Only if: (a) You want to explore infection trajectory differences between DENV and ZIKV, (b) the temporal convergence trajectory (Step 4.4) showed an interesting pattern, (c) you have experience with Monocle3 or Slingshot.

When to Skip

If temporal analysis (Step 4.4) showed flat convergence trajectory. If real timepoint data (4h-48h) already provides temporal resolution. If the interpretation would be ambiguous (pseudotime in infection context is not straightforward).

PHASE 11 — FIGURE GENERATION

Figure 1: Study Design

Purpose: Orient the reader to the three-layer framework and dataset architecture. **Inputs:** Conceptual diagram (no data analysis required). **Content:**

- Panel A: Three evidence layers (miRNAs from DENV literature, transcriptomics from both viruses, host factors from literature) shown as independent inputs
- Panel B: Discovery vs. validation dataset roles
- Panel C: Analytical workflow from data to biological model **Biological message:** This study integrates three independent evidence sources to test cross-flavivirus convergence. **Tool:** BioRender, PowerPoint, or Illustrator.

Figure 2: DENV-ZIKV Convergent Host Response

Purpose: Present the primary discovery finding. **Inputs:** DEG tables from Phase 3; merged fold-change table. **Content:**

- Panel A: Volcano plot — DENV DEGs from GSE110496
- Panel B: Volcano plot — ZIKV DEGs from GSE110496
- Panel C: UpSet plot showing DEG overlap
- Panel D: Fold-change correlation scatter plot with host factor and miRNA annotations (THE CENTRAL FIGURE from Step 4.3) **Biological message:** DENV and ZIKV produce a

quantifiably shared host transcriptomic response.

Figure 3: Multi-Layer Convergence

Purpose: Show that the shared response is enriched for proviral factors and miRNA targets. **Inputs:** Enrichment results from Phases 5-6. **Content:**

- Panel A: UpSet plot or SuperExactTest visualization of shared DEGs x proviral factors x miRNA targets
- Panel B: GSEA enrichment plot for proviral gene set
- Panel C: GSEA enrichment plot for antiviral gene set
- Panel D: Three-tier miRNA enrichment results (55 vs. 91 vs. 36 comparison table or bar plot) **Biological message:** The convergent response is functionally proviral and regulated by DENV-downregulated miRNAs.

Figure 4: Pathway Convergence and Cross-Tissue Validation

Purpose: Show functional coherence of the shared response across cell types. **Inputs:** Pathway enrichment results from Phase 7; validation pathway results from Phase 8 (immune cells and neural extension). **Content:**

- Panel A: Dot plot of enriched KEGG pathways for shared DEGs (discovery)
- Panel B: Pathway comparison heatmap across discovery, immune cell validation (GSE118305, GSE94892), and neural extension (GSE78711) showing which pathways replicate across cell types
- Panel C: Temporal convergence trajectory (correlation at each timepoint, from Step 4.4)
- Panel D (optional, if neural extension is positive): Bar chart showing replication rate of discovery shared DEGs in each validation context (macrophages, PBMCs, neural progenitor cells) **Biological message:** The shared response converges on specific biological pathways that replicate across datasets and extend from hepatoma cells to immune cells and neural progenitor cells.

Figure 5: Regulatory Network

Purpose: Show the network architecture of the shared response. **Inputs:** Cytoscape output from Phase 9. **Content:**

- Main: PPI network with color-coded proviral/antiviral genes, hub genes highlighted, three-way intersection genes with thick borders
- Inset: Hub gene ranking table **Biological message:** The shared regulatory network has identifiable hub genes and functional organization.

Figure 6: Integrative Model

Purpose: Synthesize all findings into a biological model. **Inputs:** All results. **Content:**

- Schematic: miRNA suppression → proviral gene de-repression → pathway activation → permissive state
- Key hub genes and miRNAs annotated
- Shared vs. virus-specific components indicated **Biological message:** The complete biological model of conserved flavivirus host exploitation.

Main Tables

Table 1: Dataset Summary

All datasets with accession, virus, cell type, samples, platform, role (discovery/validation), reference.

Table 2: Shared DEGs

Top 20-30 shared DEGs ranked by average fold change. Columns: gene, DENV log2FC, ZIKV log2FC, padj DENV, padj ZIKV, proviral/antiviral classification, miRNA target (Y/N), pathway membership.

Table 3: Multi-Layer Intersection Genes

All three-way intersection genes with full annotation.

Table 4: Convergent Pathways

Pathways enriched in discovery AND at least one validation dataset.

Supplementary Materials

- Table S1: Complete curated host factor list
- Table S2: Complete DENV-downregulated miRNA list with ZIKV binding status
- Table S3: Full DEG tables (all datasets including GSE78711 neural extension)
- Table S4: Complete miRNA target predictions
- Table S5: All enrichment test results (Fisher's tests, GSEA, pathway enrichment)
- Table S6: Network statistics and centrality metrics
- Table S7: Neural extension results (GSE78711 replication analysis)
- Figure S1: QC plots for GSE110496

- Figure S2: Dispersion plots and MA plots
- Figure S3: Sensitivity analysis with different DEG thresholds
- Figure S4: Full GO and Reactome enrichment results
- Figure S5: Validation DEG volcano plots (GSE118305, GSE94892, GSE78711)

PHASE 12 — FINAL INTERPRETATION

12.1: Strongest Supported Conclusion

Depends on gate outcomes:

All gates pass (G1-G6): "DENV and ZIKV converge on a shared proviral transcriptomic program regulated by miRNA-mediated post-transcriptional control. The convergent response is enriched for experimentally validated proviral host factors, targets of DENV-downregulated miRNAs with cross-flavivirus binding properties, and replicates in independent immune cell datasets across both viruses."

Gates G1-G4 pass, G5 (miRNA) fails: "DENV and ZIKV converge on a shared proviral transcriptomic program. The convergent response is enriched for proviral host factors, although the hypothesized miRNA-mediated regulation was not supported at the statistical level, suggesting alternative or additional regulatory mechanisms."

Gates G1-G3 pass, G4-G5 fail: "DENV and ZIKV converge on a shared host transcriptomic response, but the functional enrichment for proviral factors and miRNA-target regulation was not statistically significant. The convergence may reflect shared innate immune recognition rather than shared exploitation strategies."

Gates G1-G3 pass, G6 (validation) fails: "DENV and ZIKV produce a convergent host response in human hepatoma cells, but this convergence did not replicate in independent immune cell datasets, suggesting cell-type-specific rather than general flavivirus biology."

12.2: Limitations

The following limitations should be acknowledged in the Discussion:

1. GSE110496 uses a hepatoma cell line (Huh7), not primary cells. While liver cells are natural flavivirus targets, the results may not fully represent immune cell biology.
2. miRNA target predictions have inherent false positive rates. Enrichment analysis mitigates but does not eliminate this noise.
3. The host factor curated list is compiled from diverse experimental systems (different viruses, cell types, species). Some classifications may not hold in the specific cellular context studied here.
4. The study is entirely computational. All miRNA-target relationships, proviral gene

classifications, and regulatory network architectures require experimental validation.

5. Cross-dataset validation compares different cell types (Huh7 vs. macrophages vs. PBMCs vs. neural progenitor cells). Gene-level replication is expected to be partial; pathway-level replication is the appropriate metric.
6. The DENV miRNA data derives from only two published studies profiling serum miRNAs. Serotype-specific or population-specific biases may limit generalizability.
7. The neural extension dataset (GSE78711) uses ZIKV strain MR766 (African lineage, 1947 isolate), which differs from the Asian lineage strains responsible for the 2015-2016 epidemic. Transcriptomic differences between lineages have been documented. Furthermore, DENV does not infect neural progenitor cells, so the neural extension tests ZIKV-side replication only, not DENV-ZIKV convergence in neural tissue.

12.3: Testable Predictions

The model generates specific predictions for experimental follow-up:

1. Transfection of miRNA mimics for the top hub miRNAs should reduce both DENV and ZIKV replication in the same cell system.
2. siRNA knockdown of three-way intersection genes should reduce viral titers for both DENV and ZIKV.
3. The shared proviral gene set should show similar upregulation during infection with other flaviviruses (JEV, WNV).

GATE TRACKING TABLE

Use this table to record gate outcomes as the project progresses:

Gate	Phase	Test	Threshold	Result	Date	Decision
G1	3	DEGs per virus	>= 200	[pending]		
G2	4	Shared DEG overlap	Fisher $p < 0.001$	[pending]		
G3	4	Fold-change correlation	Pearson $r > 0.4$	[pending]		
G4	5	Proviral enrichment	Fisher $p < 0.05$	[pending]		
G5	6	miRNA target enrichment	Fisher $p < 0.05$ (55-set)	[pending]		
G6	8	Validation replication	Fisher $p < 0.05$ or pathway overlap	[pending]		

CRITICAL RULE: Fill this table as each gate is evaluated. Do not proceed to Phase 9 (network) or Phase 11 (figures) until Gates G1-G5 are evaluated. Do not finalize the manuscript until Gate G6 is evaluated.

END OF STANDARD OPERATING PROCEDURE